THE EXPONENT 3/4 FOR THE EXCEPTIONAL SET OF m/n, IN SHORT INTERVALS AND PROGRESSIONS

ANONYMOUS

ABSTRACT. For an integer $m \geq 4$ call n *m-exceptional* if m/n is not a sum of three unit fractions. We show that there are absolute constants $c, C > 0$ such that every interval I of length $H \geq 2$ contains at most

$$C\,H \exp\{-c(\log H)^{3/4} m^{-1/4}\}$$

m-exceptional integers, uniformly in $m \geq 4$ and in the position of I. We also give a version for arithmetic progressions, whose only loss is the smooth part of the modulus; a bound for exceptional primes in short intervals that uses no information about primes at the scale of the interval; and the density transition for primes. Uniformly in m, at most a proportion ε of the primes in $(N/2, N]$ are m-exceptional once $\log N \geq A_\varepsilon m^{1/3}$; and the proportion of m-exceptional primes tends to 1 when $\log N\,(\log\log N)^{1/3}/m^{1/3} \to 0$ (a sharpening of a first-moment count of Pomerance and Weingartner). The remaining gap is a factor $(\log\log N)^{1/3}$ in $\log N$, which is exactly the open $\log\log N$ in Elsholtz and Tao's Type I prime count. Conditionally on Selberg's eigenvalue conjecture for the groups $\Gamma_0(mdq^2)$ with even nebentypus, this loss disappears: the proportion of representable primes is $\ll (\log N)^3/m + (\log N)^2 \log^2 m/m + m^{-0.35}$, so the transition is at $\log N \asymp m^{1/3}$, with matching orders on both sides. Numerically (even $m \in [60, 300]$) half of the primes are representable at $\log N \approx 1.95\, m^{1/3}$, and we conjecture a non-degenerate profile in $\log N/m^{1/3}$. For comparison, Pomerance and Weingartner proved the exponent $2/3$ with $\varphi(m)^{-1/3}$ over initial segments, and Kandukoori, Ramesh and Kumar (unrefereed preprint) obtained the exponent $3/4$ over initial segments for $m = 4$ and for each fixed $m \geq 4$, with constants depending on m. The method is the one of an earlier internal note on $4/n$ over $[1, N]$, which we use as a black box. What we add is a local transfer to arbitrary windows, through a smooth-part decomposition, and the replacement of the numerator 4 by m, which turns cubic logarithmic class mass into $(\log X)^3/m$. **Status:** every unconditional result except the lower side of the transition is proved *relative to that note*, which is internally proved and has *not* been externally refereed; the lower side is proved relative to a published divisor-sum bound of Elsholtz and Tao and to a modification of the proof of another. The conditional sharp order is proved relative to the inputs above together with a further internal note on the Type I sum for $m = 4$ and an internal working document; the range in which it actually uses the conjecture is $m \leq (\log N)^5$. This note has not been externally refereed either. Nothing here bears on the existence of exceptions; this is not a proof of the Erdős–Straus conjecture or of Schinzel's conjecture.

1. INTRODUCTION

Let $m \geq 2$ be an integer. A positive integer n is *m-representable* if

$$\frac{m}{n} = \frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} \tag{1}$$

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has a solution in positive integers, not necessarily distinct, and *m-exceptional* otherwise. For $m \le 3$ every n is representable; for $m \ge 4$ the integer $n = 1$ is exceptional. The Erdős–Straus conjecture is the case $m = 4$ with no exception $n \ge 2$; Sierpiński conjectured the analogue for $5/n$, and Schinzel conjectured that for each $m \ge 4$ there are only finitely many m-exceptional n. For a real interval I, a modulus $q \ge 1$ and $b \in \mathbb{Z}$ put

$$E_m(I) = \#\{n \in I \cap \mathbb{Z}_{\ge 1} : n \ m\text{-exceptional}\},$$
$$E_m(I; q, b) = \#\{n \in I \cap \mathbb{Z}_{\ge 1} : n \equiv b \ (q),\ n \ m\text{-exceptional}\},$$

and $E_m(N) = E_m((0, N])$.

Background. Vaughan [V] proved $E_4(N) \ll N \exp\{-c(\log N)^{2/3}\}$, and, according to [PW, abstract], the analogous statement that most n are m-representable for each fixed m. Elsholtz and Tao [ET, §1] cite Vaughan's bound, listing other results as weaker, and Pomerance and Weingartner [PW, p. 2] describe it as the state of the art for $4/n$. Elsholtz [E, Rem. 7.3] gave an admissible constant explicit in the numerator, $E_m(N) \ll N \exp\{-c_{m,3}(\log N)^{2/3}\}$ with $c_{m,3} = 3e^{-2/3}(8m)^{-1/3} - \varepsilon$, for $N > N_m$ with N_m unspecified (O101 audit). Pomerance and Weingartner [PW, Thm. 1.3] made it uniform in the numerator: for $4 \le m \le (\log N)^2$,

$$E_m(N) \le N \exp\{-C_{\mathrm{PW}}(\log N)^{2/3}\varphi(m)^{-1/3}\}. \tag{2}$$

An internal research note [TQ] ("the note" below) proves the bound

$$E_4(N) \ll N \exp\{-c(\log N)^{3/4}\}$$

by an argument that has been internally checked but not externally refereed. The same bound, and for each *fixed* $m \ge 4$ the bound $E_m(N) \le C_m N \exp\{-c_m(\log N)^{3/4}\}$, were claimed earlier by Kandukoori, Ramesh and Kumar [KRK, Thms. 1–2] (unrefereed preprint with a Lean formalisation); we claim no priority for the exponent. All results below are proved relative to [TQ]; Section 2 lists exactly which of its statements are used.

Results. All constants $c, C, \dots$ below are absolute. As in [TQ] they are not effective, because the standard Bombieri–Vinogradov theorem is used.

Theorem A (short intervals, uniform in m)**.** *There are absolute constants $c, C > 0$ such that for every integer $m \ge 4$, every real z and every $H \ge 2$,*

$$E_m((z, z + H]) \le C\,H \exp\{-c(\log H)^{3/4}m^{-1/4}\}.$$

In particular $E_m(N) \le CN \exp\{-c(\log N)^{3/4}m^{-1/4}\}$ for all $N \ge 2$.

The bound depends only on the length of the interval. It contains the theorem of [TQ], which is the case $m = 4$, $z = 0$. The saving tends to infinity if and only if $m = o((\log H)^3)$. For $m \le \varepsilon(\log H)^3$ with $\varepsilon = \varepsilon(c, C)$ small it is a nontrivial constant factor; for larger m it is no better than the trivial bound $E_m(I) \le H + 1$.

Theorem B (progressions)**.** *There are absolute $c, C, \alpha > 0$ with the following property. Let $m \ge 4$, let I be an interval of length H, let $q \ge 1$ and $b \in \mathbb{Z}$ with $H/q \ge 2$, put $X_q = \exp\{\alpha(m \log(H/q))^{1/4}\}$, and let q_1 be the X_q-smooth part of q (the product of the prime powers $p^a \| q$ with $p \le X_q$). Then*

$$E_m(I; q, b) \le C\,q_1 \frac{H}{q} \exp\{-c(\log(H/q))^{3/4}m^{-1/4}\}.$$

Consequently $E_m(I;q,b) \le C(H/q)\exp\{-\frac{c}{2}(\log(H/q))^{3/4}m^{-1/4}\}$ *whenever* $q \le \exp\{\frac{c}{2}(\log(H/q))^{3/4}m^{-1/4}\}$, *and the factor* q_1 *is absent when every prime factor of* q *exceeds* X_q.

Corollary C (exceptional primes in short intervals)**.** *With* c, C *as in Theorem A: for* $m \ge 4$, $x \ge 3$ *and* $H \ge 2$ *with* $(\log H)^{3/4}m^{-1/4} \ge (2/c)\log\log x$,

$$\#\{p \in (x, x+H] : p \text{ } m\text{-exceptional}\} \le C\frac{H}{\log x}\exp\{-\tfrac{c}{2}(\log H)^{3/4}m^{-1/4}\}$$

(p *prime). For fixed* m *this holds for* $H \ge \exp\{C_m(\log\log x)^{4/3}\}$ *with* $C_m = (2/c)^{4/3}m^{1/3}$.

No information on primes at scale x enters: the factor $1/\log x$ is paid for by the saving. Prime-number input is needed only for relative statements (Remark 7.6).

Corollary D (the density transition, via integers)**.** *There is an absolute* C_D *such that for all* $m \ge 4$ *and all* $N \ge 16$ *with* $\log N \ge C_D m^{1/3}(\log m)^{4/3}$,

$$\#\{p \in (N/2, N] : p \text{ prime}, \ m\text{-exceptional}\} \le \frac{N}{(\log N)^2}.$$

Moreover $E_m(N) = o(N)$ *whenever* $\log N/m^{1/3} \to \infty$.

Corollary D follows from Theorem A alone; its loss $(\log m)^{4/3}$ comes from passing from integers to primes. Counting primes directly removes it. Let $\rho_{\mathrm{exc}}(m,N)$ and $\rho_{\mathrm{rep}}(m,N) = 1 - \rho_{\mathrm{exc}}(m,N)$ be the proportions of m-exceptional and m-representable primes in $(N/2, N]$, and put $A = \log N/m^{1/3}$.

Theorem U (upper side; PROVED relative to [TQ], with Bombieri–Vinogradov and Shiu)**.** *For every* $\varepsilon > 0$ *there is* A_ε (*ineffective*) *such that* $\rho_{\mathrm{exc}}(m,N) \le \varepsilon$ *for all* $m \ge 4$ *and all* N *with* $\log N \ge A_\varepsilon m^{1/3}$.

Theorem L′ (lower side; PROVED relative to [ET, Thm. 7.1] and the proof of [ET, Prop. 1.4], with Brun–Titchmarsh, Shiu and Pólya–Vinogradov)**.** *Let* $m \ge 4$, $N \ge 16$ *and* $L = \log N$. *If* $\log m \le L/10$ *and* $L \le m^{1/2}$, *then*

$$\rho_{\mathrm{rep}}(m,N) \ll \frac{(L^3 + L^2\log^2 m)\log L}{m} + m^{-0.35};$$

if $\log m > L/10$, *then* $\rho_{\mathrm{rep}}(m,N) \le \mathrm{e}^{CL/\log L}/m$. *The constants are absolute and effective. Consequently* $\rho_{\mathrm{rep}} \to 0$ *whenever* $L^3\log L/m \to 0$, *that is, whenever* $A\,(\log L)^{1/3} \to 0$.

Pomerance and Weingartner's rigorous bounds placed the transition between $\log N = m^{1/3}$ and $\log N = m^{1/2}$ [PW, p. 2]; their Poisson heuristic [PW, p. 3] predicts $m^{1/3+o(1)}$. Theorem U proves the representable half of this prediction at the exact scale $m^{1/3}$ (without a rate better than Corollary D). Theorem L′ sharpens the first-moment count in the proof of [PW, Thm. 3.1] by a factor $(m/\varphi(m))\log m$ near the transition. Together they locate the transition at $\log N \asymp m^{1/3}$ up to the factor $(\log L)^{1/3} = (\log\log N)^{1/3}$ in $\log N$ (near the transition $\log L \asymp \log m$); Corollary D with [PW] gave $(\log m)^2(m/\varphi(m))^{1/3}$. The remaining loss is exactly the Brun–Titchmarsh $\log\log N$ of [ET, Thm. 1.1], open even for $m = 4$. Conditionally

on a uniform form (SEL_m) of Selberg's eigenvalue conjecture it is removed, uniformly in m (Section 9, from [MN4]):

$$\rho_{\mathrm{rep}}(m,N) \ll \frac{L^3 + L^2 \log^2 m}{m} + m^{-0.35} \qquad (L = \log N),$$

where (SEL_m) is used only when $m \le L^5$ (Theorem 9.3). With Theorem U this places the transition at $\log N \asymp m^{1/3}$, with matching orders on both sides (Corollary 9.4). Unconditionally, for $m = 4$, the $\log\log N$ is reduced to $o(\log\log N)$ without a rate [HN, Thm. 1]; whether this transfers to general m is open (Remark 9.5). For even $m \in [60, 300]$ the computed half-point is $\log N = (1.95 \pm 0.05) m^{1/3}$, and the transition window does not shrink (EVIDENCE, Section 8.5); Conjecture C2 asserts $\rho_{\mathrm{rep}} = F(A) + o(1)$ with a non-degenerate profile F. We do not know that ρ_{exc} is monotone in N, and nothing is said about the *largest* m-exceptional integer.

Theorem A and (2) have unrelated constants (ours ineffective), so their comparison is asymptotic. The ratio of the exponents is

$$\frac{L^{3/4} m^{-1/4}}{L^{2/3}\varphi(m)^{-1/3}} = L^{1/12} \frac{\varphi(m)^{1/3}}{m^{1/4}} \gg \frac{(Lm)^{1/12}}{(\log\log 3m)^{1/3}}, \qquad L = \log N, \tag{3}$$

by $\varphi(m) \gg m/\log\log 3m$. In the range $4 \le m \le L^2$ of (2) its exponent $L^{2/3}\varphi(m)^{-1/3}$ is at least 1, which absorbs the factor C of Theorem A. Hence there is an absolute (ineffective) Λ_0 such that, for $4 \le m \le L^2$, Theorem A is stronger than (2) as soon as $(Lm)^{1/12} \ge \Lambda_0 (\log\log 3m)^{1/3}$; in particular for all such m once $N \ge N_0$. Outside this range (2) is not available. Against Elsholtz's constant $c_{m,3} \asymp m^{-1/3}$ the ratio of the exponents is exactly $(Lm)^{1/12}$, without the $\log\log$ loss, but that bound holds only for $N > N_m$ (O101 audit).

Finally (Section 10) a bound with saving in $\log x$ rather than $\log H$, of the shape $E_m((x, x+H]) \ll H \mathrm{e}^{-c(\log x)^{3/4}}$, cannot be proved for windows with $H < \mathrm{e}^{c(\log x)^{3/4}}/C$ without proving that there are no exceptions at all beyond some point (Proposition E).

Status and novelty. Every statement labelled PROVED here, except Theorem L′, the lemmas of Section 8.3 and the unconditional range of Theorem 9.3 (which follows from them), is proved *relative to* [TQ]: the steps that change are written out in full, and the remaining steps are cited from [TQ] by name. The note [TQ] is internally proved; it has *not* been refereed externally, and neither has this note. The underlying working documents [SH, MN] received three internal hostile reviews, and their repairs are incorporated. Theorem U and the Type II count come from [MN2], which received two further independent internal reviews (no fatal or major defect; minor repairs incorporated); the Type I count of Theorem L′ comes from [MN3], which replaces the divisor-sum lemma of [MN2] and received one further internal hostile review (no fatal or major defect; minor repairs incorporated). Section 9 comes from [MN4] (one further internal hostile review, no fatal or major defect) and is CONDITIONAL on (SEL_m); its proof is relative to [HN], which is internally reviewed only. Theorem L′ is proved relative to [ET, Thm. 7.1]; from the proof of [ET, Prop. 1.4] we depart by keeping a coprimality indicator and using Pólya–Vinogradov (Lemma 8.6, checked against [ET, pp. 25–32] by a reviewer, not machine-checked). Theorem U is ineffective, Theorem L′ effective.

We found no published short-interval or progression version of an exceptional-set bound for (1), for any m (Section 11); the searches were not exhaustive. We make

no novelty claim for the *method*, which is that of [TQ], and the forced classes are those of Vaughan, Elsholtz–Tao and Pomerance–Weingartner. Every forced class is an identity valid for all n in the class, and counting congruence classes is translation invariant. An expert would therefore regard the transfer to short intervals as routine for the part of the argument that concerns integers free of small prime factors. What we add is the following: a local replacement for the global semigroup step of [TQ] (Lemma 7.3); the bookkeeping that turns t^3 into t^3/m with absolute constants, with no loss of $\varphi(m)$ or $\log m$ (Sections 3–6); and the consequences above. For the transition we add a direct prime count with Bonferroni depth bounded in terms of the saving, in a reduced residue model in which every fibre is reduced (Theorem U), and, on the lower side, a sharp Type II count via the three-way split of [ET, §9] and the Pólya–Vinogradov refinement of [ET, Prop. 1.4] that [ET, Rem. 1.5] suggests, together with the coprimality gain $\varphi(k)/k$ (Theorem L′). None of this improves the exponent 3/4.

Organisation. Section 2 lists what is taken from [TQ] and isolates the single object the rest of the argument needs, a *majorant package* (Definition 2.1). Sections 3–6 construct the package for general m, writing out every step that differs from [TQ]. Section 7 contains the local transfer to arbitrary windows and progressions and proves Theorems A, B and Corollary C. Section 8 proves Corollary D and Theorems U and L′, compares them with [PW] and [E], and reports numerics and Conjecture C2; Section 9 gives the sharp order of the transition under Selberg's eigenvalue conjecture; Section 10 discusses the lower end of the window range, Section 11 the literature, and Section 12 lists open problems.

2. The black box and the majorant package

2.1. **Parameters.** We use the parameters of [TQ, (2)] with one change. Fix $\kappa = 1/480$ and the integer D of [TQ, Lemma 3.3]. For $X \geq 3$ put

$$t = \log X, \qquad K = \lfloor X^{\kappa} \rfloor, \qquad H_K = K^{10},$$

and use the blocks $I_j = (x_j, \min(2x_j, X)]$, $x_j = 2^j X^{1/2}$, with $z_j = x_j^{1/6}$; for large X one has $z_j \geq X^{1/12} > H_K^2$ [TQ, (3)]. (The note calls H_K simply H; here H is the length of a window.) The change is the multiplier set:

$$\mathcal{K}_m(K) = \{1 \leq k \leq K : (k, m) = 1\}, \qquad L_K = \operatorname{lcm} \mathcal{K}_m(K),$$

in place of $\mathcal{K}(K) = \{k \leq K : k \equiv 1 \ (4)\}$. For $\mathcal{J} \subseteq \mathcal{K}_m(K)$ put $h(\mathcal{J}) = \sum_{k \in \mathcal{J}} \varphi(k)/k^2$ and $L_{\mathcal{J}} = \operatorname{lcm} \mathcal{J}$. Throughout, $P_y = \prod_{p \leq y} p$ and $S_y(n) = \mathbf{1}_{(n, P_y) = 1}$.

2.2. **What is used from the note.** External inputs are those of [TQ, §3]: Brun–Titchmarsh in the Montgomery–Vaughan form [MV], the Bombieri–Vinogradov theorem [D, Ch. 28], and Shiu's theorem [S] (inside [TQ, Lemma 3.3]). The statements of [TQ] enter as follows.

Used verbatim.

- [TQ, Lemma 3.1] (elementary lattice sums); it is stated for every integer $1 \leq k \leq K$ and every $(c, k) = 1$.
- [TQ, Lemma 3.3] (low-ω mass), stated for every $k \leq K$, $(c, k) = 1$.
- [TQ, Lemma 4.1] (second incidence moment). It is stated for $\mathcal{J} \subseteq \mathcal{K}(K)$, but its proof uses only $\mathcal{J} \subseteq [1, K]$, so it holds for $\mathcal{J} \subseteq \mathcal{K}_m(K)$.

- The conditional-independence proof of [TQ, Theorem 6.3] (factorial moments) and [TQ, Lemma 8.1] (the even Bonferroni polynomial Q_r).
- For $m = 4$ only: [TQ, Theorem 8.2] (the assembly) as stated, together with the identity [TQ, Lemma 2.1].

Re-run with 4 **replaced by** m (each change is written out below): [TQ, Lemma 2.1] (the identity; Lemma 3.1), [TQ, Lemma 2.2] (deduplication and CRT; Lemma 3.3), [TQ, Lemma 3.2] (harmonic sum; replaced by Lemma 4.1), [TQ, Theorem 4.2, Corollary 4.3] (prime slice and fibre masses; Proposition 5.1, Corollary 5.2), [TQ, Lemma 7.1] (void; Lemma 6.1) and [TQ, Theorem 8.2] (assembly; Theorem 6.3).
Replaced. The counting step [TQ, (50)] over $[1, N]$ is replaced by its local form (Lemma 7.1), and the global transfer to all denominators [TQ, §9] (semigroup of exceptional primes and Rankin's trick) is replaced by a local smooth-part decomposition (Lemma 7.3).
Not used. The simplified route [TQ, §5], the ordered-atom replay after [TQ, Theorem 6.3] with [TQ, Lemmas 6.1, 6.2], and [TQ, §§10–11] (heuristic ceiling, checkpoints). We re-derive only the retained original route of [TQ], with the cutoff $\omega(uv) \le D \log\log X$ and congestion pruning. (The Cauchy–Schwarz route of [TQ, §5] also appears to transfer, via $\varphi(muv) \ge \varphi(m)\varphi(u)\varphi(v)$, but we have not written it out.)

The substitutions are summarised in Table 1. Where the numerator 4 occurs in [TQ] we checked it occurrence by occurrence. Apart from the unused [TQ, §5], the remark after [TQ, (21)] and [TQ, §§10–11], it appears only in statements that are re-run below: in the identity, in the multiplier condition $k \equiv 1$ (4) and [TQ, Lemma 3.2], in the deduplication bound $4uv > K$ of [TQ, Lemma 2.2], in the prime modulus $4uv$ (including the level $q \le 4x^{1/3}$ in [TQ, (8), (9)] and the main-term step of [TQ, Theorem 4.2]), in the inequality $\varphi(4uv) \ge 2\varphi(u)\varphi(v)$, and in the remark on primes $p \equiv 3$ (4) in the proof of [TQ, Lemma 7.1]. The inventory bound $|\mathcal{A}_X| \le KX^{4/3}$ does not involve the numerator. No step of the density argument (lattice sums, Shiu, congestion, moments, void, Bonferroni) looks at the quadratic character of the forced classes. So the dichotomy between $m \equiv 0 \pmod 4$ and $m \not\equiv 0 \pmod 4$ that governs *pointwise* questions for m/n [SX, §15] plays no role here.

object	note [TQ] ($m = 4$)	here (general m)
identity	$k\ell + 1 = 4uvw$	$k\ell + 1 = m\,uvw$
multipliers	$k \le K$, $k \equiv 1$ (4)	$k \le K$, $(k, m) = 1$
prime progression	$\ell \equiv -k^{-1}$ $(4uv)$	$\ell \equiv -k^{-1}$ (muv)
totient bound	$\varphi(4uv) \ge 2\varphi(u)\varphi(v)$	$\varphi(muv) \ge \varphi(m)\varphi(u)\varphi(v)$
harmonic sum	$h = \frac{2}{\pi^2}\log K + O(1)$	$h_m \asymp (\varphi(m)/m)\log K$
fibre mass	$\asymp t^3$	$\asymp s := t^3/m$
selector, depth	$y = Bt^3$, $r \approx D_B t^3$	$y = \max(2, Bs)$, $r \approx D_B s$
ledger	$\mathrm{e}^{O(t^4)}$	$\mathrm{e}^{O(ts)} = \mathrm{e}^{O(t^4/m)}$
transfer	$[1, N]$, then Rankin	any window, smooth parts

TABLE 1. The substitutions $4 \to m$ and $[1, N] \to$ window.

2.3. **The majorant package.** The transfer of Section 7 uses only the following properties.

Definition 2.1. Let $m \geq 4$, $X \geq 3$, $t = \log X$, $s \geq 1$ and $2 \leq y < X^{1/2}$. An $(m; X, s, y)$*-package with constants* (a, C_L) is a function $\nu : \mathbb{Z} \to \mathbb{R}$ together with a positive integer $\mathcal{M}$ all of whose prime factors are at most X, such that:

- (P1) $\nu(n) \geq 0$ for all n, and $\nu(n) \geq 1$ for every m-exceptional $n \geq 1$ with $(n, P_y) = 1$;
- (P2) $\nu = \sum_i \epsilon_i \mathbf{1}_{a_i \ (q_i)}$ is a finite signed combination of residue classes with every $q_i \mid \mathcal{M}$ and $T_{\mathrm{abs}} := \sum_i |\epsilon_i| \leq \mathrm{e}^{C_L t s}$;
- (P3) $\mathbb{E}_{\mathcal{M}} \nu \leq 2\mathrm{e}^{-as}$, where $\mathbb{E}_{\mathcal{M}}$ is the uniform average over $\mathbb{Z}/\mathcal{M}\mathbb{Z}$ (well defined by (P2)).

Proposition 2.2 (packages exist)**.** *There are absolute constants $a, B, C_L, s_0 > 0$ and X_a such that for every $X \geq X_a$ and every integer $4 \leq m \leq t^3$ with $s := t^3/m \geq s_0$ there is an $(m; X, s, \max(2, Bs))$-package with constants (a, C_L).*

For $m = 4$ the note gives such a package directly. Indeed [TQ, Theorem 8.2], with its own family and fixed constants, yields a $(4; X, t^3/4, B_4 t^3)$-package with constants $(4c_a, 4C_L)$ in the notation of [TQ]: property (P1) holds because every exceptional n avoids every atom [TQ, Lemma 2.2], so $H_X(n) = 0$ and $\nu_X(n) = S_y(n) Q_r(0) = 1$ when $(n, P_y) = 1$; (P2) is [TQ, (44)] with $\mathcal{M}$ from [TQ, (7)]; and (P3) is [TQ, (43)]. Note that [TQ, Theorem 8.2] states the lower bound $\nu_X \geq 1$ only for exceptional primes $n > \max(K, y)$. Its proof, however, uses only the identity, which holds for every positive integer in a forced class, so it gives (P1) as stated. The transfer of Section 7 needs only $y \leq B's + 2$ for some absolute B', which holds here with $B' = 4B_4$. Thus the case $m = 4$ of Theorems A and B follows from [TQ] and Section 7 alone. For general m, Proposition 2.2 is Theorem 6.3 below.

3. The atom family for general m

Lemma 3.1 (multiplier identity for m)**.** *Let m, k, ℓ, u, v, w be positive integers with $k\ell + 1 = m\,uvw$. If $n \geq 1$ and $nv \equiv -u \pmod{k\ell}$, then $s = (nv + u)/(k\ell)$ is a positive integer and*

$$\frac{m}{n} = \frac{1}{suw} + \frac{1}{nsvw} + \frac{1}{nuvw}.$$

In particular every positive integer in the class $-uv^{-1} \pmod{k\ell}$ is m-representable; the inverse exists because $uv \mid k\ell + 1$.

Proof. Over the common denominator $nsuvw$ the numerators sum to $nv + u + s = sk\ell + s = s(k\ell + 1) = m\,suvw$. □

For $m = 4$ this is [TQ, Lemma 2.1]. The identity was checked in exact rational arithmetic on 4821 random instances with $m \leq 60$ (`scripts/emn_identity.py`); this is a sanity check only.

Definition 3.2 (atom family)**.** Let $\mathcal{A}_X^{(m)}$ be the set of quadruples $A = (k, \ell, u, v)$ with

$$\begin{aligned} k \in \mathcal{K}_m(K), \quad \ell \in I_j \text{ prime}, \quad H_K < u, v \leq z_j, \\ (u, v) = (uv, k) = 1, \quad \omega(uv) \leq D \log\log X, \quad m\,uv \mid k\ell + 1, \end{aligned} \tag{4}$$

with atoms and atom count

$$E_A = \{n : n \equiv -uv^{-1} \pmod{k\ell}\}, \qquad H_X(n) = \sum_{A \in \mathcal{A}_X^{(m)}} \mathbf{1}_{E_A}(n).$$

The divisibility $m \mid k\ell + 1$ forces $(k, m) = (\ell, m) = 1$. Unlike [TQ], which uses $k \equiv 1 \pmod 4$ only, we admit every multiplier coprime to m and let the class $\ell \equiv -k^{-1} \pmod m$ vary with k. Restricting to $k \equiv 1 \pmod m$ would lose a factor of order $\varphi(m)$ in the multiplier harmonic sum, hence in the fibre mass. As in [TQ], the family is fixed and unpruned, and it does not depend on any residue. We use the probability space of uniform n modulo

$$\mathcal{M} = \operatorname{lcm}(L_K, P_y) \prod_{\substack{X^{1/2} < \ell \le X \\ \ell \text{ prime}}} \ell, \qquad 2 \le y < X^{1/2}, \tag{5}$$

as in [TQ, (7)]. Every prime factor of $\mathcal{M}$ is at most X.

Lemma 3.3 (deduplication and conditional independence)**.** *For all $X \ge X_0$ (absolute) and every $m \ge 4$:*

(1) *every m-exceptional integer $n \ge 1$ avoids every atom;*
(2) *at a fixed ℓ, distinct atoms have distinct projections modulo ℓ; hence the atoms of a compatible set of distinct atoms have distinct primes ℓ_i, and*

$$\mathbb{P}\Big(\bigcap_i E_{A_i}\Big) = \frac{\mathbf{1}_{\text{compatible}}}{\operatorname{lcm}(k_1, \dots, k_j) \prod_i \ell_i};$$

(3) *conditional on any residue $c \pmod{L_K}$ compatible with $S_y = 1$ (and on $S_y = 1$), the coordinates $n \bmod \ell$, $X^{1/2} < \ell \le X$, are independent and uniform, and the atom A is* active *in the fibre c, i.e. E_A meets it, precisely when $k \mid u + cv$.*

Proof. This is [TQ, Lemma 2.2]; we repeat the proof with 4 replaced by m. Part (1) is Lemma 3.1. If two atoms at the same $\ell \in I_j$ have the same projection modulo ℓ, then $\ell \mid uv' - u'v$; as $|uv' - u'v| < z_j^2 < \ell$, we get $uv' = u'v$, and reducedness gives $(u, v) = (u', v')$. Then $k\ell \equiv k'\ell \equiv -1 \pmod{m\,uv}$ and $(\ell, m\,uv) = 1$ give $k \equiv k' \pmod{m\,uv}$; since $m\,uv > H_K^2 > K$, $k = k'$. So two distinct atoms with the same ℓ are incompatible. All $\ell > X^{1/2} > K$ are coprime to L_K and to P_y, so the Chinese remainder theorem gives the intersection formula and the conditional independence; reduction modulo $k \mid L_K$ gives the activation criterion. □

No upper bound on m is needed here: $(\ell, m) = 1$ is forced by $m \mid k\ell + 1$. Lemma 3.3 was also checked numerically for $m \in \{4, 5, 6, 7, 11, 12\}$ and $\ell \in (1000, 3000)$ (19 446 atoms, no collision; `scripts/emn_identity.py`).

4. The multiplier harmonic sum

This section replaces [TQ, Lemma 3.2]. Put

$$S_m(x) = \sum_{\substack{j \le x \\ (j,m)=1}} \frac{1}{j}, \qquad h_m(K) = h(\mathcal{K}_m(K)) = \sum_{k \in \mathcal{K}_m(K)} \frac{\varphi(k)}{k^2}.$$

Lemma 4.1. *Let $m \ge 4$.*

(a) *For $x \ge m^2$: $S_m(x) \ge (\varphi(m)/m) \log x$.*
(b) *For every $K \ge 1$: $h_m(K) \ge (1 - \sum_p p^{-2}) S_m(K) \ge 0.54\, S_m(K)$.*
(c) *For every $K \ge 1$ and every prime $p \nmid m$: $\sum_{k \in \mathcal{K}_m(K),\, p \mid k} \varphi(k)/k^2 \le S_m(K)/p$.*

(d) *There is an absolute C_h such that $h_m(K) \le S_m(K) \le C_h(\varphi(m)/m)\log K$ for $K \ge \max(m,3)$.*

In particular $0.54\,(\varphi(m)/m)\log K \le h_m(K) \le C_h(\varphi(m)/m)\log K$ for $K \ge m^2$.

Proof. (a) By Möbius inversion over $e \mid m$, $S_m(x) = \sum_{e\mid m}\mu(e)e^{-1}\sum_{i\le x/e} i^{-1}$. For real $y \ge 1$ one has $\sum_{i\le y} i^{-1} = \log y + \gamma + \theta_y/y$ with $|\theta_y| \le 1$, and here $x/e \ge x/m \ge m$. Hence

$$S_m(x) = \frac{\varphi(m)}{m}(\log x + \gamma) - \sum_{e\mid m}\frac{\mu(e)\log e}{e} + R, \qquad |R| \le \frac{2^{\omega(m)}}{x}.$$

Differentiating $f(\sigma) = \sum_{e\mid m}\mu(e)e^{-\sigma} = \prod_{p\mid m}(1-p^{-\sigma})$ at $\sigma = 1$ gives $-\sum_{e\mid m}\mu(e)\log e/e = f'(1) = (\varphi(m)/m)\sum_{p\mid m}\log p/(p-1) \ge 0$. Finally $2^{\omega(m)}/x \le m/m^2 = 1/m < \gamma\varphi(m)/m$, because $\varphi(m) \ge 2$ and $2\gamma > 1$.

(b) Since $\varphi(k)/k = \prod_{p\mid k}(1-1/p) \ge 1 - \sum_{p\mid k} 1/p$, and $k \in \mathcal{K}_m(K)$ with $p \mid k$ means $k = pj$ with $p \nmid m$ and $j \in \mathcal{K}_m(K/p)$,

$$\begin{aligned} h_m(K) &\ge S_m(K) - \sum_{p\nmid m}\frac{1}{p}\sum_{\substack{k\in\mathcal{K}_m(K)\\ p\mid k}}\frac{1}{k} \\ &= S_m(K) - \sum_{p\nmid m}\frac{S_m(K/p)}{p^2} \ge \Big(1 - \sum_p p^{-2}\Big)S_m(K), \end{aligned}$$

and $\sum_p p^{-2} < 0.4523$.

(c) Write $k = pj$ as in (b) and use $\varphi(k)/k^2 \le 1/k$.

(d) The first inequality is $\varphi(k) \le k$. Every $j \le K$ coprime to m has all its prime factors in $\{p \le K : p \nmid m\}$, so

$$S_m(K) \le \prod_{\substack{p\le K\\ p\nmid m}}\Big(1-\frac{1}{p}\Big)^{-1} = \prod_{p\le K}\Big(1-\frac{1}{p}\Big)^{-1}\prod_{\substack{p\mid m\\ p\le K}}\Big(1-\frac{1}{p}\Big).$$

For $K \ge m$ every prime factor of m is at most K, so the last product is $\varphi(m)/m$. Mertens' theorem bounds the first by an absolute multiple of $\log K$ for $K \ge 3$. □

A numerical check of (a)–(c) for $m = 4,\dots,60$ and $m = 210$ up to $K = 10^5$ found no failure, with $S_m(K)/((\varphi(m)/m)\log K) \in [1.05, 1.23]$ (`scripts/emn_hm.py`). For $m = 4$ the note's sub-progression $k \equiv 1 \pmod 4$ gives $(2/\pi^2)\log K + O(1)$ [TQ, Lemma 3.2], a constant fraction of h_4.

5. Fibre masses for general m

For $k \ge 1$, $(c,k) = 1$ and $1 \le H_K < z$ let, as in [TQ, §3],

$$\mathcal{S}(k,c;H_K,z) = \{(u,v) : H_K < u, v \le z,\ (u,v) = (v,k) = 1,\ u \equiv -cv\ (k)\}$$

(so $(u,k) = 1$ automatically), and for $\mathcal{J} \subseteq \mathcal{K}_m(K)$ and $(c, L_{\mathcal{J}}) = 1$ let $r_{\mathcal{J}}(u,v;c) = \#\{k \in \mathcal{J} : k \mid u + cv\}$. For $\mathcal{J} \subseteq \mathcal{K}_m(K)$ and a residue c let $f_{c;\mathcal{J}}(\ell)$ be the number of atoms $A \in \mathcal{A}_X^{(m)}$ with $\ell_A = \ell$, $k_A \in \mathcal{J}$ and $k_A \mid u_A + cv_A$.

Proposition 5.1 (pruned prime slice for m)**.** *There are absolute $a_h, A_h > 0$ and X_h such that for every $X \ge X_h$, every integer $4 \le m \le t^3$, every $\mathcal{J}$ with $1 \in \mathcal{J} \subseteq \mathcal{K}_m(K)$*

and every $(c, L_{\mathcal{J}}) = 1$ *the following holds. In each block* I_j *keep only the triples* (k, u, v) *with*

$$\begin{aligned} (u,v) \in \mathcal{S}(k, c; H_K, z_j), \quad k \in \mathcal{J}, \\ \omega(uv) \le D \log\log X, \quad r_{\mathcal{J}}(u,v;c) \le T_X := t^4, \end{aligned} \tag{6}$$

and let $f^{\mathrm{good}}_{c;\mathcal{J}}(\ell)$ *be the number of such triples with* $m\,uv \mid k\ell + 1$, *for* $\ell \in I_j$ *prime. Then*

$$a_h \frac{t^2 h(\mathcal{J})}{\varphi(m)} \le \sum_{X^{1/2} < \ell \le X} \frac{f^{\mathrm{good}}_{c;\mathcal{J}}(\ell)}{\ell} \le \sum_{X^{1/2} < \ell \le X} \frac{f_{c;\mathcal{J}}(\ell)}{\ell} \le A_h \frac{t^2 h(\mathcal{J})}{\varphi(m)}.$$

Proof. We follow the proof of [TQ, Theorem 4.2]; the changes are the modulus $m\,uv$ in place of $4uv$ and the resulting factor $1/\varphi(m)$. Each counted triple (k, u, v) together with ℓ is an atom of $\mathcal{A}_X^{(m)}$, active in the fibre c. By Lemma 3.3(2) distinct such atoms at the same ℓ have distinct classes modulo ℓ, so $f^{\mathrm{good}}_{c;\mathcal{J}} \le f_{c;\mathcal{J}}$, and both count distinct classes. Fix a full block, $x = x_j$, $z = z_j = x^{1/6}$, $\Lambda = \log(z/H_K) \ge \frac{1}{2}\log z$.

Harmonic mass. Summing [TQ, Lemma 3.3] over $k \in \mathcal{J}$ (its cutoff $D\log\log z$ is at most $D\log\log X$) gives $\sum_{k\in\mathcal{J}} \sum 1/(uv) \ge \frac{1}{8}\Lambda^2 h(\mathcal{J}) \ge \frac{1}{32}(\log z)^2 h(\mathcal{J})$, the inner sum over $(u, v) \in \mathcal{S}(k, c; H_K, z)$ with $\omega(uv) \le D\log\log X$. The triples removed by $r_{\mathcal{J}} > T_X$ carry harmonic mass at most

$$\sum_{r_{\mathcal{J}}(u,v;c) > T_X} \frac{r_{\mathcal{J}}(u,v;c)}{uv} \le \frac{1}{t^4} \sum_{\substack{H_K < u,v \le z \\ (u,v)=1}} \frac{r_{\mathcal{J}}(u,v;c)^2}{uv} \le \frac{C_{\mathrm{cg}} (\log z)^2 (1+\kappa t)^3}{t^4}$$

by [TQ, Lemma 4.1] (whose constant is called C_2 there), which applies to $\mathcal{J} \subseteq [1, K]$. Since $h(\mathcal{J}) \ge 1$ (as $1 \in \mathcal{J}$), the retained harmonic mass is at least $\frac{1}{64}(\log z)^2 h(\mathcal{J})$ for large X.

Main term. A retained triple asks for primes $\ell \equiv -k^{-1} \pmod q$, $q = m\,uv$; this class is reduced because $(k, m) = (k, uv) = 1$. As $m \le t^3$ and $uv \le x^{1/3}$, $q \le t^3 x^{1/3}$. Using $\mathrm{li}(2x) - \mathrm{li}(x) \ge x/\log(2x)$ and $\varphi(m\,uv) \le \varphi(m)\,uv$, the total main-term weight of the retained triples is at least

$$\frac{x}{\varphi(m)\log 2x} \sum_{\text{retained}} \frac{1}{uv} \ge a_1 \frac{x \log x\, h(\mathcal{J})}{\varphi(m)} \qquad (a_1 > 0 \text{ absolute}).$$

Bombieri–Vinogradov. For a fixed q, the quotient $uv = q/m$ is fixed, it has at most $2^{\omega(uv)} \le (\log X)^{D\log 2}$ ordered coprime splittings (u, v), and each splitting has at most T_X retained multipliers. So the retained multiplicity of q is $W(q) \le t^{4 + D\log 2}$, exactly as in [TQ, (26)]. Fix $R = D\log 2 + 17$ in [TQ, (9)]. As $x \ge X^{1/2}$ we have $t \le 2\log x$, and $q \le t^3 x^{1/3}$ is below the level $x^{1/2}(\log x)^{-A_R}$ for large X. Grouping the retained triples by q, the total error is therefore at most $(2\log x)^{4+D\log 2} C_R x (\log x)^{-R} \ll x(\log x)^{-13}$. The main term is at least $a_1 x \log x\, t^{-3} \gg x(\log x)^{-2}$, by $h(\mathcal{J}) \ge 1$ and $\varphi(m) \le m \le t^3$. *This is the only place where the range* $m \le t^3$ *is used for the lower bound*; any range $m \le t^A$ would do with R depending on A. Hence the retained triples have at least $\frac{a_1}{2} x \log x\, h(\mathcal{J})/\varphi(m)$ prime incidences $\ell \in (x, 2x]$, all distinct classes, and dividing by $2x$ gives $\sum_{\ell\in(x,2x]} f^{\mathrm{good}}_{c;\mathcal{J}}(\ell)/\ell \ge \frac{a_1}{4} \log x\, h(\mathcal{J})/\varphi(m)$. Summing over the $\asymp t$ full blocks, with $\log x \asymp t$, proves the lower bound.

Upper bound. Drop the pruning and the cutoff. For $(u, v) \in \mathcal{S}(k, c; H_K, z)$, Brun–Titchmarsh [TQ, (8)] with $V = 2x$ and $q = m\,uv \le x^{0.34}$, together with

$\varphi(m\,uv) \geq \varphi(m)\varphi(u)\varphi(v)$ (valid since $\varphi(ab) \geq \varphi(a)\varphi(b)$ for all a, b), gives at most $8x/(\varphi(m)\varphi(u)\varphi(v)\log x)$ primes $\ell \in (x, 2x]$. By [TQ, (12)] the sum over $k \in \mathcal{J}$ and (u, v) is $\ll x \log x\, h(\mathcal{J})/\varphi(m)$. Divide by x and sum over the blocks, enlarging the last one. □

Corollary 5.2 (uniform fibre masses)**.** *There are absolute $a_u, C_u > 0$ such that for $X \geq X_h$, every integer $4 \leq m \leq t^3$ and* every *residue $c \pmod{L_K}$, with $\mathcal{J}_c = \{k \in \mathcal{K}_m(K) : (k, c) = 1\}$, $f_c = f_{c;\mathcal{J}_c}$ and $\mu_c = \sum_\ell f_c(\ell)/\ell$,*

$$a_h \frac{t^2 h(\mathcal{J}_c)}{\varphi(m)} \leq \mu_c \leq A_h \frac{t^2 h(\mathcal{J}_c)}{\varphi(m)} \leq C_u \frac{t^3}{m}.$$

If $(c, L_K) = 1$, then moreover $\mu_c \geq a_u t^3/m$.

Proof. As in [TQ, Corollary 4.3]. An atom active in the fibre c has $c \equiv -uv^{-1} \pmod{k}$, a unit, so $k \in \mathcal{J}_c$; thus f_c counts all active atoms, $1 \in \mathcal{J}_c$ and $(c, L_{\mathcal{J}_c}) = 1$, and Proposition 5.1 applies to $\mathcal{J} = \mathcal{J}_c$. For large X, $K \geq t^6 \geq m^2$ and $\kappa t/2 \leq \log K \leq \kappa t$. Lemma 4.1(d) gives $h(\mathcal{J}_c) \leq h_m(K) \leq C_h \kappa t\, \varphi(m)/m$, hence the upper bound with $C_u = A_h C_h \kappa$. If $(c, L_K) = 1$, then $\mathcal{J}_c = \mathcal{K}_m(K)$ and Lemma 4.1(a),(b) give $h(\mathcal{J}_c) \geq 0.27\, \kappa t\, \varphi(m)/m$, so $a_u = 0.27\, \kappa a_h$ works. □

So on reduced fibres the mass is $\asymp t^3/m$ with absolute constants, and on every fibre it is $\ll t^3/m$; for instance $c = 0$ has $\mathcal{J}_0 = \{1\}$ and $\mu_0 \asymp t^2/\varphi(m)$. The factor $1/\varphi(m)$ from the prime progression modulo $m\,uv$ and the factor $\varphi(m)/m$ in h_m cancel to $1/m$, so nothing is lost from $m/\varphi(m)$. A deduplicated toy computation (`scripts/emn_mass2.py`; $\ell \in (10^6, 2 \cdot 10^6]$, $K = 170$, $u, v \in (7, 60]$, two reduced fibres) gave $m\mu_c \in [0.98, 1.13]$ for $m = 4, \ldots, 13$, while $\varphi(m)\mu_c$ ranged over $[0.33, 1.04]$. This is EVIDENCE for the scaling only; the corollary does not use it.

6. Void, moments and the assembly for general m

Throughout this section $4 \leq m \leq t^3$ and

$$s = t^3/m \geq 1, \qquad y = \max(2, Bs), \qquad S_y(n) = \mathbf{1}_{(n, P_y) = 1},$$

with an absolute B fixed in Lemma 6.1. For large X, $y \leq \max(2, Bt^3/4) < K < X^{1/2}$.

Lemma 6.1 (conditioned void for m)**.** *There are absolute $a_v, B, s_0 > 0$ and X_v such that for $X \geq X_v$, $4 \leq m \leq t^3$ and $s \geq s_0$,*

$$\mathbb{P}(H_X = 0 \mid S_y = 1) \leq \mathrm{e}^{-a_v s}.$$

Proof. We re-run [TQ, Lemma 7.1]. Reveal $c \equiv n \pmod{L_K}$, keeping the condition $S_y = 1$, and put

$$Z(c) = \sum_{\substack{y < p \leq K,\ p \mid L_K \\ p \mid c}} \frac{1}{p}.$$

For the supported primes $p > y$ the events $p \mid c$ are independent with probabilities $1/p$, also given $S_y = 1$, and the computation [TQ, (39)–(40)] gives $\mathbb{E}(\mathrm{e}^{yZ} \mid S_y = 1) \leq \exp\{(\mathrm{e} - 1) y \sum_{p > y} p^{-2}\} \leq \mathrm{e}^{Cz}$ with C_Z absolute; it uses only $y/p < 1$, so it holds for every $y \geq 2$. Hence $\mathbb{P}(Z > 1/4 \mid S_y = 1) \leq \mathrm{e}^{-y/4 + C_Z}$.

If $p \mid L_K$ then $p \mid c$ if and only if $p \mid n$; so on $S_y = 1$ no prime $p \le y$ dividing L_K divides c. A multiplier $k \in \mathcal{K}_m(K) \setminus \mathcal{J}_c$ is divisible by some prime $p \mid (c, L_K)$, and such p satisfy $p > y$ and $p \nmid m$. By Lemma 4.1(b),(c),

$$h(\mathcal{J}_c) \ge h_m(K) - \sum_{\substack{y<p\le K,\ p\mid L_K \\ p\mid c}} \frac{S_m(K)}{p} \ge (0.54 - Z(c))S_m(K).$$

On $Z(c) \le 1/4$, Lemma 4.1(a) (as $K \ge m^2$) and $\log K \ge \kappa t/2$ give $h(\mathcal{J}_c) \ge 0.29\,(\varphi(m)/m)\log K \ge 0.145\,\kappa t\,\varphi(m)/m$. By Lemma 3.3(3), in the fibre c the event $H_X = 0$ asks that each coordinate $n \bmod \ell$ avoid $f_c(\ell)$ distinct classes, independently in ℓ. With Corollary 5.2,

$$\begin{aligned}\mathbb{P}(H_X = 0 \mid c, S_y = 1) &= \prod_{\ell}\Big(1 - \frac{f_c(\ell)}{\ell}\Big) \\ &\le \mathrm{e}^{-\mu_c} \le \exp\Big\{-a_h\,0.145\,\kappa\frac{t^3}{m}\Big\} = \mathrm{e}^{-2a_v s},\end{aligned}$$

with $a_v := 0.0725\,\kappa a_h$. Put $B = 8a_v$. For $s \ge 2/B$ we have $y = Bs$, and the bad fibres have probability at most $\mathrm{e}^{-2a_v s + C_Z}$. Averaging over c, $\mathbb{P}(H_X = 0 \mid S_y = 1) \le (1 + \mathrm{e}^{C_Z})\mathrm{e}^{-2a_v s} \le \mathrm{e}^{-a_v s}$ for $s \ge s_0 := \max(2/B, \log(1+\mathrm{e}^{C_Z})/a_v)$. □

The threshold for Z is the absolute number $1/4$: the union-bound loss $Z \cdot S_m(K)$ is measured against $h_m \ge 0.54\,S_m$ in the same units, so no dependence on m enters. In [TQ] the corresponding constant is called η.

Lemma 6.2 (factorial moments)**.** *For $X \ge X_h$, $4 \le m \le t^3$, every real $2 \le y < X^{1/2}$ and every integer $j \ge 1$, $\mathbb{E}((H_X)_j \mid S_y = 1) \le (C_u s)^j$, where $(h)_j = h(h-1)\cdots(h-j+1)$.*

Proof. The conditional-independence proof of [TQ, Theorem 6.3] applies verbatim. In a fibre c (with $S_y = 1$, which conditions no coordinate $n \bmod \ell$ as $y < X^{1/2}$), Lemma 3.3 shows that H_X is a sum of independent Bernoulli variables with means $f_c(\ell)/\ell$, at most one atom being hit at each ℓ. So $\mathbb{E}((H_X)_j \mid c, S_y = 1) = \sum_{\ell_1,\dots,\ell_j \text{ distinct}} \prod_i f_c(\ell_i)/\ell_i \le \mu_c^j$, and $\mu_c \le C_u s$ for *every* c by Corollary 5.2. □

Theorem 6.3 (assembly for m)**.** *Let a_v, B, s_0 be as in Lemma 6.1 and put $D_B = \max(\mathrm{e}^2 C_u, a_v)$ and $C_L = B + 3D_B + 8$. There is an absolute X_a such that for $X \ge X_a$, $4 \le m \le t^3$ and $s = t^3/m \ge s_0$, with r the least even integer $\ge D_B s$, the function*

$$\nu_X(n) = S_y(n)\,Q_r(H_X(n)), \qquad Q_r(h) = \sum_{i=0}^{r}(-1)^i\binom{h}{i},$$

together with $\mathcal{M}$ from (5), is an $(m; X, s, y)$-package with constants (a_v, C_L). In particular Proposition 2.2 holds.

Proof. (P1) By [TQ, Lemma 8.1], $Q_r(0) = 1$ and $Q_r(h) \ge 0$ for all integers $h \ge 0$; so $\nu_X \ge 0$. If n is m-exceptional with $(n, P_y) = 1$, then $H_X(n) = 0$ by Lemma 3.3(1), and $\nu_X(n) = 1$.

(P3) By [TQ, Lemma 8.1], $Q_r(h) \le \mathbf{1}_{h=0} + \binom{h}{r+1}$. By Lemmas 6.1 and 6.2 and $(r+1)! \ge ((r+1)/\mathrm{e})^{r+1}$,

$$\mathbb{E}(Q_r(H_X) \mid S_y = 1) \le \mathrm{e}^{-a_v s} + \Big(\frac{\mathrm{e}C_u s}{r+1}\Big)^{r+1} \le \mathrm{e}^{-a_v s} + \mathrm{e}^{-(r+1)} \le 2\mathrm{e}^{-a_v s},$$

since $r+1 \ge D_B s \ge \mathrm{e}^2 C_u s$ and $D_B \ge a_v$. Multiply by $\mathbb{P}(S_y = 1) \le 1$.

(P2) For all integers n, $\binom{H_X(n)}{i} = \sum_{\mathcal{B}\subseteq\mathcal{A}_X^{(m)},\,|\mathcal{B}|=i} \prod_{A\in\mathcal{B}} \mathbf{1}_{E_A}(n)$ and $S_y(n) = \sum_{d\mid P_y} \mu(d)\mathbf{1}_{d\mid n}$. Expanding, each term has coefficient ± 1 and is empty or a single class modulo $\operatorname{lcm}(d, k_1\ell_1, \dots, k_i\ell_i)$, which divides $\mathcal{M}$. Choosing $k \le K$, $\ell \le X$ and $u, v \le X^{1/6}$ shows $|\mathcal{A}_X^{(m)}| \le KX^{4/3} \le \mathrm{e}^{(4/3+\kappa)t}$, as in [TQ, (48)]. Hence, as in [TQ, (49)],

$$\log T_{\mathrm{abs}} \le \pi(y)\log 2 + \log(r+1) + r(4/3+\kappa)t.$$

Using $s, t \ge 1$: $\pi(y)\log 2 \le y \le Bs + 2 \le (B+2)ts$; $\log(r+1) \le r \le D_B s + 2 \le (D_B+2)ts$; and $r(4/3+\kappa)t \le 2(D_B+2)ts$. So $\log T_{\mathrm{abs}} \le C_L ts$. All prime factors of $\mathcal{M}$ are at most X, and $2 \le y < X^{1/2}$. □

The whole dependence on m is the substitution $t^3 \to s = t^3/m$ in the saving, the selector and the Bonferroni depth. The cost $(1+\kappa)t$ of a single atom modulus does not change, which is why the ledger is $\mathrm{e}^{O(ts)} = \mathrm{e}^{O(t^4/m)}$ and not $\mathrm{e}^{O(s)}$.

7. Local transfer to windows and progressions

In [TQ] the majorant is counted on $[1, N]$ in exactly one place, [TQ, (50)], using only $\#\{n \le N : n \equiv a\ (q)\} = N/q + O(1)$ with error at most one. Nothing in that step uses the left endpoint. The prime-number inputs (BV, BT, Shiu) live at the scale $X = \mathrm{e}^t$ and serve only to construct the package. The one step of [TQ] that is not local is the passage from primes to all integers in [TQ, §9], through the semigroup generated by exceptional primes and Rankin's trick. We replace it by Lemma 7.3.

Lemma 7.1 (local mean, class by class)**.** *Let $\nu = \sum_i \epsilon_i \mathbf{1}_{a_i\ (q_i)}$ be a finite signed combination of residue classes with every $q_i \mid \mathcal{M}$ and $T_{\mathrm{abs}} = \sum_i |\epsilon_i|$. Let $Q \ge 1$, $\beta \in \mathbb{Z}$, and let $I = (z, z+H]$ be any real interval, $H > 0$. Then*

$$\sum_{\substack{n\in I\cap\mathbb{Z}\\ n\equiv\beta\ (Q)}} \nu(n) = H\,\mathbb{E}_{\mathbb{Z}/\operatorname{lcm}(\mathcal{M},Q)}\big[\nu\,\mathbf{1}_{\beta\ (Q)}\big] + \theta\, T_{\mathrm{abs}}, \qquad |\theta| \le 1.$$

Proof. For each i the set $\{n \equiv a_i\ (q_i)\} \cap \{n \equiv \beta\ (Q)\}$ is empty or a single class modulo $q_i' = \operatorname{lcm}(q_i, Q)$. In the second case it has density $1/q_i'$ in $\mathbb{Z}/\operatorname{lcm}(\mathcal{M}, Q)$, and every half-open real interval of length H contains $H/q_i' + \theta_i$ of its elements, with $|\theta_i| \le 1$. Multiply by ϵ_i and sum over i. □

Lemma 7.2 (large-prime moduli are independent)**.** *Let $\nu \ge 0$ be $\mathcal{M}$-periodic, and let $Q = Q_1Q_2$ with $(Q_1, Q_2) = (Q_2, \mathcal{M}) = 1$. Then for every β, $\mathbb{E}_{\mathbb{Z}/\operatorname{lcm}(\mathcal{M},Q)}[\nu\,\mathbf{1}_{\beta\ (Q)}] \le \mathbb{E}_{\mathcal{M}}\nu/Q_2$.*

Proof. By the Chinese remainder theorem $\mathbb{Z}/\operatorname{lcm}(\mathcal{M}, Q) \cong \mathbb{Z}/\operatorname{lcm}(\mathcal{M}, Q_1) \times \mathbb{Z}/Q_2$. The function ν depends only on the first coordinate, with mean $\mathbb{E}_{\mathcal{M}}\nu$ there, and $\mathbf{1}_{\beta\ (Q)} \le \mathbf{1}_{\beta\ (Q_2)}$, which depends only on the second coordinate and has mean $1/Q_2$. Use $\nu \ge 0$. □

Lemma 7.3 (smooth-part decomposition)**.** *Let $m \ge 4$ and $y \ge 2$. Write each integer $n \ge 1$ uniquely as $n = dn'$ with $P^+(d) \le y$ and $(n', P_y) = 1$, where $P^+(d)$ is the largest prime factor of d ($P^+(1) = 1$).*

(a) *If n is m-exceptional, then so is n'; here $n' = 1$ is allowed and is m-exceptional.*

(b) *If $d > D_0 \ge 1$, then d has a divisor $d' \in (D_0, yD_0]$ with $P^+(d') \le y$.*
(c) *For $D_0 \ge 1$, $\sum_{d'>D_0,\ P^+(d')\le y} 1/d' \le D_0^{-1/2}\mathrm{e}^{4\sqrt{y}}$.*

Proof. (a) If $m/n' = \sum_i 1/x_i$, then $m/n = \sum_i 1/(dx_i)$. The integer 1 is m-exceptional because three unit fractions sum to at most $3 < m$. (b) Write $d = p_1 \cdots p_r$ with primes $p_i \le y$ and let i be least with $p_1 \cdots p_i > D_0$; then $d' = p_1 \cdots p_i \le yD_0$. (c) By Rankin's trick the sum is at most $D_0^{-1/2} \prod_{p\le y}(1-p^{-1/2})^{-1}$. By convexity, $-\log(1-u) \le 2u$ for $0 \le u \le 2^{-1/2}$, and $\sum_{p\le y} p^{-1/2} \le \sum_{j\le y} j^{-1/2} \le 2\sqrt{y}$. □

Part (a) is the divisor closure used in [TQ, §9]; we apply it to the smooth part only, which keeps everything inside the window.

Proposition 7.4 (transfer)**.** *Let $\nu, \mathcal{M}$ be an $(m; X, s, y)$-package with constants (a, C_L), and put $D_0 = \mathrm{e}^{as}$. Let I be an interval of length $H > 0$, let $q \ge 1$, $b \in \mathbb{Z}$, and let q_2 be the largest divisor of q all of whose prime factors exceed X. Then*

$$E_m(I; q, b) \le \frac{H}{q_2}\Big(\mathrm{e}^{-as/2+4\sqrt{y}} + 2(1+as)\mathrm{e}^{-as}\Big) + yD_0 + D_0\mathrm{e}^{C_L ts}.$$

Proof. Split the m-exceptional $n \in I$ with $n \equiv b\ (q)$ according to their y-smooth part d (Lemma 7.3). Since d is y-smooth and $y < X$, $(d, q_2) = 1$.

Case $d > D_0$. By Lemma 7.3(b), $d' \mid n$ for some y-smooth $d' \in (D_0, yD_0]$. For fixed d', the $n \in I$ with $d' \mid n$ and $n \equiv b\ (q)$ form at most one class modulo $\mathrm{lcm}(d', q)$, a multiple of $d'q_2$; there are at most $H/(d'q_2) + 1$ of them. Summing over d' and using Lemma 7.3(c) gives at most $(H/q_2)D_0^{-1/2}\mathrm{e}^{4\sqrt{y}} + yD_0$.

Case $d \le D_0$. Then $n' = n/d$ lies in I/d, an interval of length H/d; it is m-exceptional and coprime to P_y, so $\nu(n') \ge 1$ by (P1). The congruence $dn' \equiv b\ (q)$ is either insoluble or confines n' to one class β_d modulo $Q_d = q/(d, q)$. As $(d, q_2) = 1$, $Q_d = Q_{d,1}q_2$ with $(Q_{d,1}, q_2) = (q_2, \mathcal{M}) = 1$. By Lemmas 7.1 and 7.2 and (P2), (P3), the number of such n is at most

$$\sum_{\substack{n' \in I/d \\ n' \equiv \beta_d\ (Q_d)}} \nu(n') \le \frac{H}{d} \cdot \frac{2\mathrm{e}^{-as}}{q_2} + \mathrm{e}^{C_L ts}.$$

Summing over $d \le D_0$ with $\sum_{d\le D_0} 1/d \le 1 + \log D_0 = 1 + as$ gives at most $2(H/q_2)(1+as)\mathrm{e}^{-as} + D_0\mathrm{e}^{C_L ts}$. □

If I contains integers $n \le 0$, they are simply not counted on the left; on the right, $\nu \ge 0$ makes the count over all integers of I/d an upper bound. Every bound depends on I only through H.

Proof of Theorem A and Theorem B. Take a, B, C_L, s_0, X_a from Theorem 6.3, so that $y = \max(2, Bs) \le Bs + 2$. (For $m = 4$ one may instead use [TQ] directly, as explained after Proposition 2.2.) Fix s_1 with $4\sqrt{Bs+2} \le as/4$ for $s \ge s_1$. For $s \ge \max(s_0, s_1)$, Proposition 7.4 gives

$$E_m(I; q, b) \le C_1 \frac{H}{q_2}\mathrm{e}^{-as/4} + \mathrm{e}^{C_2 ts}, \tag{7}$$

with C_1 absolute and $C_2 = C_L + a + B + 3$; for the second term use $y \le (B+2)s$, $D_0 = \mathrm{e}^{as}$ and $s, t \ge 1$.

Let $\mathcal{H} = \log(H/q)$, which is $\log H$ for Theorem A (where $q = 1$). Choose X through

$$t^4 = \frac{m\mathcal{H}}{2C_2}, \qquad \text{so that} \qquad \mathrm{e}^{C_2 ts} = \mathrm{e}^{C_2 t^4/m} = (H/q)^{1/2}$$

and

$$s = \frac{t^3}{m} = \Big(\frac{\mathcal{H}}{2C_2}\Big)^{3/4} m^{-1/4}.$$

This is $X = X_q$ with $\alpha = (2C_2)^{-1/4}$ in Theorem B. The side conditions are: $X \ge X_a$; $m \le t^3$, i.e. $s \ge 1$; and $s \ge \max(s_0, s_1)$. Since $t^3 = ms \ge 4s$, the first follows from $s \ge (\log X_a)^3/4$. Choose an absolute s_* at least all these thresholds and so large that $s \ge s_*$ implies $as/4 \le \mathcal{H}/2$; this is possible since $s \le \mathcal{H}^{3/4}$ once $2C_2 \ge 1$.

If $s \ge s_*$, then $(H/q)^{1/2} \le (H/q)\mathrm{e}^{-as/4}$ and $H/q_2 = q_1 H/q$, so (7) gives $E_m(I; q, b) \le (C_1 + 1)q_1(H/q)\mathrm{e}^{-as/4}$, which is the claim with $c = (a/4)(2C_2)^{-3/4}$. If $s < s_*$, then $\mathcal{H}^{3/4} m^{-1/4} < C_3 := (2C_2)^{3/4} s_*$, and the trivial bound $E_m(I; q, b) \le H/q + 1 \le 2H/q \le 2\mathrm{e}^{cC_3}(H/q)\mathrm{e}^{-c\mathcal{H}^{3/4}m^{-1/4}}$ suffices. Take $C = \max(C_1 + 1, 2\mathrm{e}^{cC_3})$. The two consequences in Theorem B follow from $q_1 \le q$ and from $q_1 = 1$. □

The parameter X is a real number; the package exists for every real $X \ge X_a$, so the continuous choice of t is legitimate. The proof does not use the semigroup step of [TQ, §9], so for $m = 4$, $I = (0, N]$ it gives a second derivation of the integer statement of [TQ, Theorem 1.1] from its prime-free core [TQ, Theorem 8.2].

Remark 7.5 (smooth moduli and prime moduli)**.** The factor q_1 comes from the crude bound $\mathbf{1}_{\beta\ (Q)} \le \mathbf{1}_{\beta\ (q_2)}$ in Lemma 7.2. Conditioning on $n \bmod q_1$ fixes n modulo primes of L_K and P_y, that is, it partially fixes the fibre variable c of Lemma 6.1, and it may do so adversarially: the void bound is an average over c in which bad fibres have probability $\mathrm{e}^{-\Omega(y)}$. For example, for $q_1 = \prod_{y<p\le w} p$ and $b \equiv 0\ (q_1)$ the void lemma's *bound* gives nothing, since $Z(c) \ge \sum_{y<p\le w,\, p\mid L_K} 1/p$ may exceed $1/4$. We have *not* shown that the conditional mean of ν_X is then large. Such progressions are harmless for the equation itself: every $n \equiv 0\ (q_1)$ is representable as soon as one prime $p \mid q_1$ is. For $m = 4$, every prime modulus q with H/q large is covered without the factor q_1 (with saving constant $c/2$): if $q \le X_q$ then $q \le \exp\{\frac{c}{2}(\log(H/q))^{3/4}4^{-1/4}\}$ for large H/q. For general m this absorption fails, because $\log X_q/((\log(H/q))^{3/4}m^{-1/4}) \asymp (m/\log(H/q))^{1/2}$ need not be bounded. Primes q with $\exp\{\frac{c}{2}(\log(H/q))^{3/4}m^{-1/4}\} < q \le X_q$ are not covered by Theorem B.

Proof of Corollary C. Put $\mathcal{S} = (\log H)^{3/4}m^{-1/4}$. Theorem A bounds the count by $CH\mathrm{e}^{-c\mathcal{S}} = CH\mathrm{e}^{-c\mathcal{S}/2}\mathrm{e}^{-c\mathcal{S}/2}$, and $\mathrm{e}^{-c\mathcal{S}/2} \le 1/\log x$ by hypothesis. □

In the same way Theorem B gives, for primes $p \equiv b\ (q)$ in $(x, x + H]$, the bound $C(H/(q\log x))\exp\{-\frac{c}{4}(\log(H/q))^{3/4}m^{-1/4}\}$ whenever $q \le \exp\{\frac{c}{2}(\log(H/q))^{3/4}m^{-1/4}\}$ and $(\log(H/q))^{3/4}m^{-1/4} \ge (4/c)\log\log x$. Below the range of Corollary C, Theorem A still saves an unbounded factor over H (for fixed m and, say, $H = (\log x)^A$), but not enough to absorb the normalisation $1/\log x$.

Remark 7.6 (relative statements need prime input)**.** To bound the *proportion* of exceptional primes in $(x, x + H]$ one needs a lower bound for $\pi(x + H) - \pi(x)$. For instance, Baker, Harman and Pintz [BHP] show that $(u - u^{0.525}, u]$ contains $\gg u^{0.525}/\log u$ primes for $u \ge u_0$; we quote this from the literature and did not recheck the exact form of the statement. For $2x^{0.525} \le H \le x$, tile $(x, x + H]$ greedily

from $u = x + H$ downward by $\lfloor H/(2x)^{0.525} \rfloor \geq 1$ disjoint intervals $(u - u^{0.525}, u]$. This gives $\pi(x+H) - \pi(x) \gg H/\log x$, and with Theorem A the proportion of m-exceptional primes in $(x, x+H]$ is $\ll (\log x)\exp\{-c'(\log x)^{3/4}m^{-1/4}\}$, with $c' = 0.525^{3/4}c$. This conclusion is CONDITIONAL on the quoted statement of [BHP], which we could not check against the source. It is $\ll \exp\{-\frac{c'}{2}(\log x)^{3/4}m^{-1/4}\}$ once $(\log x)^{3/4}m^{-1/4} \geq (2/c')\log\log x$. Almost-all versions for shorter H, and versions for primes in progressions, follow in the same way from the corresponding prime-counting results; we have not checked their ranges.

8. The density transition

Throughout this section $L = \log N$, $A = L/m^{1/3}$, $\pi^*(N) = \pi(N) - \pi(N/2)$, and $\pi^*(N; q, a)$ counts the primes $p \in (N/2, N]$ with $p \equiv a\ (q)$. Let $\rho_{\mathrm{exc}}(m, N)$ and $\rho_{\mathrm{rep}}(m, N) = 1 - \rho_{\mathrm{exc}}(m, N)$ be the proportions of m-exceptional and m-representable primes in $(N/2, N]$. The working documents for Theorems U and L′ are [MN2] and [MN3].

8.1. Corollary D: integers first.

Proof of Corollary D. Apply Theorem A to $I = (N/2, N]$ and put $L = \log(N/2)$; the count is at most $CN\exp\{-cL^{3/4}m^{-1/4}\}$. Let $f(L) = cL^{3/4}m^{-1/4} - 2\log L$; then $f'(L) > 0$ for $L > (8/(3c))^{4/3}m^{1/3}$. Let $L_0 = Am^{1/3}(\log m)^{4/3}$ with $A \geq (8/(3c))^{4/3}(\log 4)^{-4/3}$, so that f increases on $[L_0, \infty)$. Since $\log\log m \leq \log m$,

$$\begin{aligned} f(L_0) &= cA^{3/4}\log m - 2\log A - \tfrac{2}{3}\log m - \tfrac{8}{3}\log\log m \\ &\geq \Big(cA^{3/4} - \tfrac{10}{3} - \tfrac{2\log A}{\log 4}\Big)\log m, \end{aligned}$$

which is at least $\log(2C)$ for A large enough (absolute), as $\log m \geq \log 4$. Put $C_D = 2A$. If $\log N \geq C_D m^{1/3}(\log m)^{4/3}$, then $L \geq \frac{1}{2}\log N \geq L_0$ (as $\log N \geq 2\log 2$), so the count is at most $CN\mathrm{e}^{-f(L)}L^{-2} \leq N/(2L^2) \leq N/(\log N)^2$; the last step uses $\sqrt{2}\log(N/2) \geq \log N$, which holds for $N \geq 16$. This is $o(\pi(N) - \pi(N/2))$. For integers, the saving in Theorem A is $(\log N)^{3/4}m^{-1/4} = (\log N/m^{1/3})^{3/4}$. □

The loss $(\log m)^{4/3}$ arises because Theorem A counts integers: primes have density $1/L$, so the integer saving $\mathrm{e}^{-cA^{3/4}}$ says something about primes only once it beats $1/L$, i.e. once $A^{3/4} \gg \log L \asymp \log m$ near the transition. To see the transition at the scale $A \asymp 1$ we count primes directly, with Bonferroni depth *bounded* in terms of the required saving; then the moduli stay below $N^{0.45}$ and the Bombieri–Vinogradov theorem suffices.

8.2. The upper side: Theorem U.

We use the atom family $\mathcal{A}_X^{(m)}$ of Definition 3.2, $t = \log X$, $s = t^3/m$, $H = H_X$ and the even Bonferroni polynomial Q_r of Theorem 6.3; recall $\mathbf{1}_{h=0} \leq Q_r(h) \leq \mathbf{1}_{h=0} + \binom{h}{r+1}$ for integers $h \geq 0$ [TQ, Lemma 8.1]. No selector S_y is used.

Lemma 8.1 (reduced model)**.** *Let $\mathcal{M}' = L_K \prod_{X^{1/2} < \ell \leq X} \ell$ and let $\tilde{n}$ be uniform on $(\mathbb{Z}/\mathcal{M}'\mathbb{Z})^\times$. For $X \geq X_h$, $4 \leq m \leq t^3$ and every even $r \geq 2\mathrm{e}^2 C_u s$,*

$$\mathbb{E}\, Q_r(H(\tilde{n})) \leq \mathrm{e}^{-a_u s} + \mathrm{e}^{-(r+1)},$$

with a_u, C_u from Corollary 5.2.

Proof. By the Chinese remainder theorem for unit groups, $c = \tilde{n} \bmod L_K$ is uniform on the reduced classes and the coordinates $\tilde{n} \bmod \ell$ are uniform on $(\mathbb{Z}/\ell\mathbb{Z})^\times$, all independent ($\ell > X^{1/2} > K$, so $\ell \nmid L_K$). Given c, the atom A meets the fibre iff $k \mid u + cv$ (Lemma 3.3(3)), and the active atoms at a fixed ℓ have pairwise distinct classes modulo ℓ (Lemma 3.3(2)), all non-zero since $0 < u < \ell$. So, given c, $H = \sum_\ell \xi_\ell$ with independent $\xi_\ell \sim \mathrm{Bernoulli}(f_c(\ell)/(\ell-1))$, whence $\mathbb{P}(H = 0 \mid c) \le \mathrm{e}^{-\mu_c}$ and $\mathbb{E}(\binom{H}{r+1} \mid c) \le (2\mu_c)^{r+1}/(r+1)!$, using $\ell/(\ell-1) \le 2$. Every fibre is reduced, so $a_u s \le \mu_c \le C_u s$ by Corollary 5.2, and $(r+1)! \ge ((r+1)/\mathrm{e})^{r+1}$ gives $\mathbb{E}(Q_r(H) \mid c) \le \mathrm{e}^{-a_u s} + (2\mathrm{e}C_u s/(r+1))^{r+1} \le \mathrm{e}^{-a_u s} + \mathrm{e}^{-(r+1)}$. Average over c. □

Compared with Lemma 6.1, neither the selector nor the bad-fibre event $Z(c) > 1/4$ is needed: in the reduced model every fibre is reduced, which is exactly the situation for primes. Since both $a_u s$ and $C_u s$ bound μ_c on the nonempty set of reduced fibres, $a_u \le C_u$.

Expand $Q_r(H(n)) = \sum_{\mathcal{B}} (-1)^{|\mathcal{B}|} \mathbf{1}[n \in \bigcap_{A\in\mathcal{B}} E_A]$ over $\mathcal{B} \subseteq \mathcal{A}_X^{(m)}$ with $|\mathcal{B}| \le r$, dropping the $\mathcal{B}$ with empty intersection. By Lemma 3.3(2) each remaining $\mathcal{B}$ has distinct primes ℓ_A, and $\bigcap_{\mathcal{B}} E_A$ is one unit class $a_{\mathcal{B}}$ modulo $q_{\mathcal{B}} = q'_{\mathcal{B}} \prod_{A\in\mathcal{B}} \ell_A$, where $q'_{\mathcal{B}} = \mathrm{lcm}(k_A) \le K^r$. Let $M(q) = \#\{\mathcal{B} : q_{\mathcal{B}} = q\}$.

Lemma 8.2 (multiplicity of moduli)**.** *For fixed r and $X \ge X_h$, $S_2 := \sum_q M(q)^2/\varphi(q) \le C(r)\, t^{17r+4^r}$, with $C(r)$ depending only on r (and the fixed κ).*

Proof. As $k \le K < X^{1/2} < \ell$, the primes ℓ_A of $\mathcal{B}$ are exactly the prime factors of q above $X^{1/2}$, each carrying one atom, which is fixed by a divisor $k \le K$ of $q' = q/\prod \ell$ and a pair $u, v \mid k\ell + 1$. So $M(q) \le \prod_{\ell \mid q,\, \ell > X^{1/2}} g_{q'}(\ell)$ with $g_{q'}(\ell) = \sum_{k \mid q',\, k \le K} \tau(k\ell+1)^2$, and $g_{q'}(\ell)^2 \le \tau(q') \sum_{k\mid q',\, k\le K} \tau(k\ell+1)^4$. Shiu's theorem [S, Thm. 1] for τ^4 on the progression $1 \bmod k$ in $(kx, 2kx]$, $x \ge X^{1/2} \ge k^2$, gives $\sum_{x<\ell\le 2x} \tau(k\ell+1)^4 \ll (k/\varphi(k))\, x\, t^{15}$; summing dyadically and using $k/\varphi(k) \ll \log t$, $\sum_\ell \tau(k\ell+1)^4/(\ell-1) \ll t^{17}$. Hence $\sum_\ell g_{q'}(\ell)^2/(\ell-1) \le C\tau(q')^2 t^{17}$. Since q' and the ℓ's are coprime and φ is multiplicative,

$$S_2 \le \sum_{q' \le K^r} \frac{1}{\varphi(q')} \sum_{j \le r} \frac{(C\tau(q')^2 t^{17})^j}{j!} \le (r+1) C^r t^{17r} \sum_{q' \le K^r} \frac{\tau(q')^{2r}}{\varphi(q')},$$

and $\sum_{n\le x} \tau(n)^{2r}/\varphi(n) \le \prod_{p\le x}(1 + \sum_{\nu\ge 1}(\nu+1)^{2r}/\varphi(p^\nu)) \ll_r (\log x)^{4^r}$, with $\log K^r \le r\kappa t$. □

Theorem 8.3. *There are absolute $a, C_1, s_0 > 0$ and for each $s \ge s_0$ an (ineffective) $N_0(s)$ such that for all $m \ge 4$ and $N \ge N_0(s)$ with $\log N \ge C_1 s^{4/3} m^{1/3}$,*

$$\#\{p \in (N/2, N] : p \text{ } m\text{-exceptional}\} \le 3\mathrm{e}^{-as}\pi^*(N).$$

Proof. Take $t = (sm)^{1/3}$, $X = \mathrm{e}^t$ and r the least even integer $\ge 2\mathrm{e}^2 C_u s$. Then $m \le t^3$, and $X \ge X_h$ once s_0 is large, as $t \ge (4s)^{1/3}$. All moduli satisfy $q_{\mathcal{B}} \le (KX)^r = \mathrm{e}^{(1+\kappa)rt}$, and $(1+\kappa)rt \le (1+\kappa)(2\mathrm{e}^2 C_u s + 2)(sm)^{1/3} \le 0.45L$ for C_1 large; so $q_{\mathcal{B}} \le N^{0.45}$. An m-exceptional prime p has $H(p) = 0$ (Lemma 3.3(1)), hence $Q_r(H(p)) = 1$, so

$$\#\{\text{exc. } p\} \le \sum_{N/2 < p \le N} Q_r(H(p)) = \sum_{\mathcal{B}} (-1)^{|\mathcal{B}|} \pi^*(N; q_{\mathcal{B}}, a_{\mathcal{B}}).$$

Write $\pi^*(N; q, a) = \pi^*(N)/\varphi(q) + \Delta(q, a)$. As $\mathbb{P}(\tilde{n} \equiv a_{\mathcal{B}}\ (q_{\mathcal{B}})) = 1/\varphi(q_{\mathcal{B}})$, the main terms sum exactly to $\pi^*(N)\,\mathbb{E}\,Q_r(H(\tilde{n})) \le \pi^*(N)(\mathrm{e}^{-a_u s} + \mathrm{e}^{-(r+1)})$ by

Lemma 8.1. For the errors, Brun–Titchmarsh gives $|\Delta(q,a)| \le CN/(\varphi(q)L)$ for $q \le N^{0.45}$, and the Bombieri–Vinogradov theorem (applied at N and $N/2$) gives $\sum_{q\le N^{0.45}} \max_{(a,q)=1} |\Delta(q,a)| \ll_{A'} NL^{-A'}$; recentring from $(\operatorname{li} N - \operatorname{li}(N/2))/\varphi(q)$ to $\pi^*(N)/\varphi(q)$ costs $\ll N\mathrm{e}^{-c\sqrt{L}} \sum_q M(q)/\varphi(q) \ll N\mathrm{e}^{-c\sqrt{L}}(S_2 \sum_{q\le N} 1/\varphi(q))^{1/2}$ by the prime number theorem. By Cauchy–Schwarz and Lemma 8.2,

$$\sum_{\mathcal{B}} |\Delta(q_{\mathcal{B}}, a_{\mathcal{B}})| \le \Big(\sum_q M(q)^2 \frac{CN}{\varphi(q)L}\Big)^{1/2} \big(C_{A'} NL^{-A'}\big)^{1/2} \le C'(r)\frac{N}{L^3}$$

for $A' = 17r + 4^r + 5$, using $t \le L$. For $N \ge N_0(s)$ this and the recentring cost are at most $N/L^2 \le \mathrm{e}^{-a_u s}\pi^*(N)$, as $\pi^*(N) \ge N/(3L)$. Finally $\mathrm{e}^{-(r+1)} \le \mathrm{e}^{-a_u s}$ since $r \ge 2\mathrm{e}^2 C_u s \ge a_u s$; take $a = a_u$. □

Proof of Theorem U. Fix $s \ge s_0$ with $3\mathrm{e}^{-as} \le \varepsilon$, and put

$$A_\varepsilon = \max(C_1 s^{4/3}, \log N_0(s)).$$

If $\log N \ge A_\varepsilon m^{1/3}$, then also $\log N \ge A_\varepsilon$, so both hypotheses of Theorem 8.3 hold. □

Quantitatively, Theorem 8.3 gives $\rho_{\mathrm{exc}} \le 3\exp\{-a(A/C_1)^{3/4}\}$ once N is large in terms of A; the dependence of N_0 on s comes from the Bombieri–Vinogradov constant at level $L^{-A'(r)}$, $r \asymp s$. Theorem U does *not* improve the rate $\rho_{\mathrm{exc}} \ll 1/L$ of Corollary D: that would need $s \asymp \log L$, hence depth $r \asymp \log L$, outside the fixed-A' range of Bombieri–Vinogradov. The dependence on m enters only through Corollary 5.2 ($\mu_c \asymp t^3/m$ on every reduced fibre), the scaling that the lower side has to match.

8.3. **The lower side: Theorem L′.** Let $m \ge 4$ and p prime. Every solution of $m/p = 1/x + 1/y + 1/z$ is of Type I or Type II, and [PW, Cor. 2.2, Cor. 2.4]: p has a Type I solution iff there are $a, d, f \in \mathbb{N}$ with $f \mid ma^2d + 1$ and $mad \mid p + f$ (then $mad \le 3p$ [PW, Lemma 7.4]); p has a Type II solution iff there are $a, b, e \in \mathbb{N}$ with $e \mid a + b$ and $mab \mid p + e$ (and then $(a, b) = 1$ may be assumed). Pomerance and Weingartner bound the primes covered by these congruences by first moments with Brun–Titchmarsh; their losses relative to the heuristic L^3/m are (a) in Type II, Brun–Titchmarsh near modulus N and $\varphi(m)$ in place of m, and (b) in Type I, the factor $\log(1 + k)$ of [ET, Prop. 1.4], $\varphi(m)$ in place of m, and Brun–Titchmarsh near modulus N. Below, (a) is removed; in (b) the factor $m/\varphi(m)$ is removed by a coprimality gain in [ET, Prop. 1.4] (Lemma 8.6) and the $\log(1 + k)$ is confined to a lower-order range; the Brun–Titchmarsh loss in Type I is not removed. It is exactly the $\log\log N$ in the prime bound $\sum_{p\le N} f_I(p) \ll N\log^2 N \log\log N$ of [ET, Thm. 1.1], which is open even for $m = 4$: Elsholtz and Tao attribute it to Brun–Titchmarsh for very short progressions and conjecture that it can be removed (discussion after [ET, Thm. 1.1]; see also [ET, §9], whose parenthetical says "Type II" where Type I is evidently meant). In this subsection every $\ll$ has an absolute constant, and $S'_m(E) = \sum_{e\le E,\,(e,m)=1} 1/\varphi(e) \ll (\varphi(m)/m)\log E$ for $E \ge \log m$ (by the Euler product, as in Lemma 4.1).

Lemma 8.4 (two harmonic sums)**.** (a) *For $Y \ge \max(2, \log m)$:*

$$\sum_{\substack{e\le 2Y\\(e,m)=1}} \sum_{\substack{a,b\le Y,\ (a,b)=1\\ e\mid a+b}} \frac{1}{\varphi(a)\varphi(b)} \ll \frac{\varphi(m)}{m}\log^3 Y + \log^2 Y.$$

(b) *For $U \ge 2$ with $\log U \ge \frac12 \log(mU)$:* $\sum_{u\le U} \tau_3(u)/\varphi(mu-1) \ll (\log^3 U + m^{0.02})/m$.

Proof. (a) As $(a,b)=1$, $(e,a)=1$. With $b/\varphi(b) = \sum_{g\mid b} \mu^2(g)/\varphi(g)$, for $(\beta,e)=1$ and $\beta_0 \in [1,e]$ the least representative, $\sum_{b\le Y,\, b\equiv\beta\,(e)} 1/\varphi(b) \ll 1/\varphi(\beta_0) + (\log Y)/e$. Summing over a in classes $a_0 \bmod e$ and using $\sum_{a_0+\beta_0=e} 1/(\varphi(a_0)\varphi(\beta_0)) \ll (\log 2e)^2/e$ gives $\ll (\log^2 Y)/e$ per $e \ge 2$, and $\log^2 Y$ for $e=1$; sum over $e \le 2Y$ with $S'_m(2Y)$.

(b) Put $n = mu-1$ and $y = \log(mU)$. As $\omega(n) \le 2y$, $n/\varphi(n) \le \mathrm{e}^4 \sum_{s\mid n,\, s\mid P_y} \mu^2(s)/\varphi(s)$. By Rankin's trick with $\sigma = 1/\log y$, $\sum_{s\mid P_y,\, s>U^{1/2}} 1/\varphi(s) \ll U^{-\sigma/2}(\log y)^{\mathrm{e}} \ll y^{-10}$, so the $s > U^{1/2}$ contribute $\ll 1/m$. For $s \le U^{1/2}$, $(s,m)=1$, the u lie in the class $m^{-1} \bmod s$. The $u > s^{1.1}$ are handled by Shiu's theorem for τ_3 on dyadic ranges (modulus $s \le u^{1/1.1}$), giving $\ll \log^3 U/m$ after summing over s. For $u \le s^{1.1}$ swap the sums: the total is $\le \sum_u \tau_3(u)\tau(mu-1)(mu-1)^{-1} \max_{s\ge u^{1/1.1}} \varphi(s)^{-1} \ll m^{0.02}/m$. □

Proposition 8.5 (Type II)**.** *For $\log m \le L/10$, the number of primes $p \in (N/2, N]$ with a Type II solution is $\ll (N/L)(L^3 + m^{0.02})/m$.*

Proof. Write $p+e = mabd$, $a+b = ce$, $(a,b)=1$, and take $a \le b$ by symmetry. As $(e,m) \mid p$, $(e,m)=1$. Following [ET, §9], with $abd \le 2N/m$, $(made)(macd)(mab)^{1/2} \le m^{5/2}a^2b \cdot ce \cdot d^2 \le 8m^{1/2}N^2$, so one of the three moduli is $\le 3m^{1/5}N^{4/5} \le 3N^{0.82}$, and Brun–Titchmarsh applies with $\log(N/q) \gg L$. (i) If $made \le 3N^{0.82}$, fix (a,d,e); then p lies in one class modulo $made$ (note $(e,ad)=1$: $(e,a) \mid (a,b)=1$, and $(e,d) \mid p$ with $e \le a+b < p$), and the count is $\ll (N/(L\varphi(m)))L^2 S'_m(N) \ll NL^2/m$. (ii) If $macd \le 3N^{0.82}$, fix (a,c,d); then $p = (macd-1)e - ma^2d$ lies in one class modulo $mu-1$, $u = acd$, and Lemma 8.4(b) with $U = 3N^{0.82}/m$ gives $\ll (N/L)(L^3 + m^{0.02})/m$. (iii) If $mab \le 3N^{0.82}$, fix (a,b,e); the class is $-e \bmod mab$, and Lemma 8.4(a) gives $\ll (N/(L\varphi(m)))((\varphi(m)/m)L^3 + L^2) \ll NL^2/m$, using $\varphi(m) \gg m/\log\log m$ and $\log\log m \le L$. □

Lemma 8.6 ([ET, Prop. 1.4] with the coprimality gain)**.** *Fix $l \ge 1$. Let $k \ge 1$, $A, B \ge 2$, and $\Lambda = 1 + \min\bigl(\log(1+k),\, k^{1/6}\log(2kA)/A^{1/2}\bigr)$.* (a) *If $kB^2 \le A^l$ and $A \ge \omega(k)+2$, then $\sum_{a\le A,\, b\le B} \tau(kab^2+1) \ll_l (\varphi(k)/k)AB\log A$.* (b) *If $kA \le B^l$ and $B \ge \omega(2k)+2$, then $\sum_{a\le A,\, b\le B} \tau(kab^2+1) \ll_l (\varphi(k)/k)AB\log B\cdot\Lambda$.*

Proof. Put $W_K(y) = \sum_{n\le y,\,(n,K)=1} 1/n$. Every n counted has all its prime factors $\le y$ and none dividing K, so by Mertens $W_K(y) \le \prod_{p\le y,\, p\nmid K}(1-1/p)^{-1} \ll (\varphi(K)/K)\log y \cdot \mathrm{e}^{2\omega(K)/y}$, which is $\ll (\varphi(K)/K)\log y$ for $y \ge \omega(K)$.

(a) For fixed b apply [ET, Thm. 7.1] to $P(a) = kb^2a+1$ (coefficients $\le A^l$, at most one root modulo every prime power, and none modulo q unless $(q,kb)=1$): $\sum_{a\le A} \tau(P(a)) \ll_l A\, W_k(A)$. Sum over b.

(b) For fixed a, [ET, Thm. 7.1] for $P(b) = kab^2+1$ (degree 2, coefficients $\le B^l$, $\rho(p^j) \le 4$ [ET, p. 30]) gives $\sum_{b\le B} \tau(P(b)) \ll_l B\sum_{q\le B} \rho(q)/q \le 9B\sum_{q_0\le B \text{ odd}} \rho(q_0)/q_0$. For odd q_0,

$$\rho(q_0) \le \mathbf{1}[(q_0,k)=1] \sum_{r\mid q_0} \Big(\frac{-ka}{r}\Big).$$

If $(q_0,ka)=1$ this is [ET, p. 31]; if $(q_0,k)=1<(q_0,a)$ the left side is 0 and the right side is a product of non-negative local factors; if $(q_0,k)>1$ both sides vanish. The indicator is the only new input: [ET] drop it, and this is what costs the factor $k/\varphi(k)$. Writing $q_0=rn$ with $(n,2k)=1$, $\sum_{q_0}\rho(q_0)/q_0\le\sum_{r\le B\text{ odd},\,(r,k)=1}(\frac{-ka}{r})W(B/r)/r$ with $W=W_{2k}$, and $r\mapsto W(B/r)/r$ is positive and decreasing. Sum over $a\le A$ and split the r-range at $1\le Q_1\le Q_2\le B$. Squares $r=s^2\le Q_2$ contribute $\le AW(B)\sum_s s^{-2}\ll AW(B)$. For non-squares $r\le Q_1$, $\sum_{a\le A}(\frac{-ka}{r})=(\frac{-k}{r})\sum_{a\le A}(\frac{a}{r})$ is a sum of a non-principal character modulo r, so Pólya–Vinogradov (valid for imprimitive characters with $\ll\sqrt{r}\log r$) gives the contribution $\ll W(B)\sqrt{Q_1}\log Q_1$. Non-squares in $(Q_1,Q_2]$ contribute trivially $\le AW(B)(1+\log(Q_2/Q_1))$. All $r>Q_2$ (square terms are non-negative and were already bounded above): for each a the Kronecker symbol of $-4ka$ is a non-principal real character modulo $4ka$ which equals $(\frac{-ka}{r})$ on odd r, so Pólya–Vinogradov and partial summation give $\ll\sqrt{ka}\log(2ka)\,W(B)/Q_2$, in total $\ll W(B)A^{3/2}k^{1/2}\log(2kA)/Q_2$. The choice $Q_1=\min(B,A^2/\log^2(2A))$, $Q_2=\min(B,\max(Q_1,kA))$ gives $\ll AW(B)(1+\log(1+k))$, and $Q_1=Q_2=\min(B,Ak^{1/3})$ gives $\ll W(B)(A+A^{1/2}k^{1/6}\log(2kA))$. So the sum over a is $\ll AW(B)\Lambda$, and $W_{2k}(B)\ll(\varphi(k)/k)\log B$. □

Remark 8.7. [ET, Prop. 1.4] bounds the same sum by $AB\log(A+B)\log(1+k)$. Its proof bounds the symbol by 1 in a middle range of r and drops the indicator $(q_0,k)=1$; [ET, Rem. 1.5] suggests the first repair. The square r need a separate range: there the character is principal (a harmless slip on [ET, p. 31]). For $q\equiv1\ (k)$ all divisors of q are coprime to k, so the gain $\varphi(k)/k$ has the right order (EVIDENCE: for $A=B=120$ and $k=4,12,\dots,60060$ the normalised sum divided by $\varphi(k)/k$ stays in $[0.81,1.12]$ [MN3]).

Proposition 8.8 (Type I)**.** *For $\log m\le L/10$ and $L\le m^{1/2}$, the number of primes $p\in(N/2,N]$ with a Type I solution is $\ll(N/L)\big[(L^3+L^2\log^2 m)\log L/m+m^{-0.35}\big]$.*

Proof. For $m\le m_0$ (an absolute constant fixed below) the bound is trivial, as the count is $\le N\le(N/L)\,m_0L^3\log L/m$. Let $m>m_0$. By Brun–Titchmarsh the count is [PW, (3.2)]

$$\ll\sum_{mad\le3N}\frac{\tau(ma^2d+1)\,N}{\varphi(m)\varphi(ad)\log(2+N/mad)}.$$

By [ET, (A.12)], $1/\varphi(ad)\ll(ad)^{-1}\sum_{s|a,\,t|d}1/(st)$; put $a=sa'$, $d=td'$, $k=ms^2t\ge m$, so $ma^2d+1=ka'^2d'+1$, and split each block $ad\in(X,2X]$ into $\ll L$ boxes $a'\sim A'$, $d'\sim D'$, $A'D'\asymp X/(st)$; a box with $A'<2$ or $D'<2$ is enlarged to $A',D'\ge2$ by positivity. Take $l=30$: then $kA'^2\le D'^l$ if $D'\ge\max(A',k^{1/28})$ and $kD'\le A'^l$ if $A'>\max(D',k^{1/28})$; choose m_0 with $k^{1/28}\ge\omega(2k)+2$ for $k\ge m_0$. *Boxes with* $\max(A',D')\ge k^{1/28}$: Lemma 8.6, with linear variable d' and quadratic variable a' (part (a) if $D'\ge A'$, (b) if $D'<A'$), gives $\ll(\varphi(k)/k)(X/st)\log X\cdot\Lambda$, where $\Lambda\le2$ unless $D'<k^{1/3}\log^2(2kX)$; at most $\ll\log(2k)+\log L$ boxes per block are of that kind, each with $\Lambda\le1+\log(1+k)$. As $\varphi(k)/k\le\varphi(m)/m$, the weighted block sum is $\ll(\varphi(m)/m)\sum_{s,t}(st)^{-2}[L^2+L\log^2(2ms^2tL)]\ll(\varphi(m)/m)(L^2+L\log^2 m)$, using $\log L\le\log m$. *Boxes with* $\max(A',D')<k^{1/28}$: here $n=ka'^2d'+1\le8k^{1.11}$ and $\tau(n)\ll k^{0.012}$. They occur only if $X\le4st\,k^{1/14}\le4m^{1/14}(st)^{8/7}$; grouping the pairs (s,t) by $u=st$ (at most $\tau(u)$ pairs), their weighted mass is $\ll m^{0.013}\min\big(1,(X/m^{0.072})^{-0.85}\big)$, which

summed over the blocks (Brun–Titchmarsh weight ≤ 1) is $\ll m^{0.014}$: a count $\ll Nm^{0.014}/\varphi(m) \ll (N/L)m^{-0.35}$, using $L \le m^{1/2}$. Summing the first kind over the blocks $X = 3N2^{-j}/m$, $j \le 2L$, with Brun–Titchmarsh weight $\ll 1/j$ gives $\ll (N/\varphi(m))(\varphi(m)/m)(L^2 + L\log^2 m)\log L$. □

Lemma 8.9 (very large m)**.** *If $\log m > L/10$, then $\rho_{\mathrm{rep}} \le \mathrm{e}^{CL/\log L}/m$.*

Proof. Count classes trivially: $\#\{n \le N : n \equiv r \ (q)\} \le N/q + 1$ and $\tau(n) \le \mathrm{e}^{CL/\log L}$ for $n \le 9N^2$. The Type I classes give $\sum_{mad\le 3N} \tau(ma^2d+1)(N/(mad)+1) \ll \mathrm{e}^{CL/\log L}(N/m)L^2$, and Type II likewise; then divide by $\pi^*(N) \gg N/L$. □

Proof of Theorem L′. The first range is Propositions 8.5 and 8.8 with $\pi^*(N) \gg N/L$ and $m^{0.02}/m \le m^{-0.35}$; the second is Lemma 8.9. For the consequence: if $\rho_{\mathrm{rep}} > 0$, some $p \le N$ has a solution, so $m \le 3p$ and $L \ge \log(m/3)$; so $L^3\log L/m \to 0$ forces $m \to \infty$, and then $L^2\log^2 m \log L/m \le (L^3 + \log^6 m)\log L/m \to 0$ and $m^{-0.35} \to 0$, while $L \le m^{1/2}$ holds eventually. In the range $\log m > L/10$ the bound of Lemma 8.9 is $\le m^{10C/\log L - 1} \to 0$, as $L \ge \log(m/3) \to \infty$. □

8.4. **The gap, and comparison with earlier bounds.** Theorem U gives $\rho_{\mathrm{exc}} \to 0$ as $A \to \infty$, uniformly in $m \ge 4$. For Theorem L′ note that $L^3\log L/m = A^3\log L$; hence $\rho_{\mathrm{rep}} \to 0$ whenever $A(\log L)^{1/3} \to 0$. So the transition lies at $\log N \asymp m^{1/3}$ up to a factor

$$(\log L)^{1/3} = (\log\log N)^{1/3}$$

in $\log N$; near the transition $\log L = \frac{1}{3}\log m + O(1)$, so this is a constant times $(\log m)^{1/3}$, uniformly in the arithmetic of m. Corollary D together with the proof of [PW, Thm. 3.1] gave the factor $(\log m)^2(m/\varphi(m))^{1/3}$, and the version of Theorem L′ in [MN2], with $\varphi(m)$ in place of m, gave $(m\log m/\varphi(m))^{1/3}$. The remaining loss has a single source, on the lower side: the Brun–Titchmarsh $\log L$ in the Type I count, which is the open $\log\log N$ of [ET, Thm. 1.1]. Removing it gives the exact scale: $\rho_{\mathrm{rep}} \to 0$ if $A \to 0$ and $\rho_{\mathrm{exc}} \to 0$ if $A \to \infty$. This is done conditionally on Selberg's eigenvalue conjecture in Section 9.

Remark 8.10 (a conditional route to the exact scale)**.** For $m = 4$, an internal working document [TTL] proves $\sum_{p\le N} f_I(p) \ll N\log^2 N$, i.e. removes the $\log\log N$ of [ET, Thm. 1.1], CONDITIONALLY on Selberg's eigenvalue conjecture for $\Gamma_0(4dq^2)$ with even nebentypus, uniformly in the level ([HN, Thm. 8.2], from [TTL, Thm. 8.1]; recorded as (D)32 in the campaign ledger `DISCOVERIES.md`, after two rounds of internal hostile review). The method writes $ef - 4a^2d = 1$ as a matrix in $\mathrm{SL}_2(\mathbb{Z})$, so that for fixed d the Type I solutions are level-d Heegner forms of discriminant $-4d$ with uniformly separated roots, and replaces Brun–Titchmarsh in the bad range by a spectral large sieve for box Poincaré series; with only the Kim–Sarnak bound it leaves a strip of positive width (a level-averaged variant reduces the loss for $m = 4$ unconditionally to $o(\log\log N)$, [HN, Thm. 1]). The m-uniform version of the conditional argument, for $m \le L^5$, is carried out in [MN4]: it removes the factor $\log L$ from Theorem L′ and gives the exact scale $\log N \asymp m^{1/3}$, conditionally on Selberg's conjecture (Section 9). The argument removes the primes $\ell \mid m$ from the sieving set and pays for this with the gain $\varphi(m)/m$ of the masses [MN3, remarks after Thm. 3.8].

Pomerance–Weingartner. Their proof of [PW, Thm. 3.1] counts, for primes $p \le N$, the ones covered by Type I congruences, $\ll (N/\varphi(m))\log^2 N \log\log N \log m$,

which they simplify to $(N/\varphi(m))\log^2 N\log^2 m$ for $\mathrm{e}^{m^{1/4}} \ll N < \mathrm{e}^m$, and the ones covered by Type II congruences, $\ll (N/\varphi(m))\log^2 N\log\log N$ [PW, pp. 6–8]. The proof is written for one $N = N(m)$, with $\varphi(m)/\log^2 m = C\log^3 N$, but the counts hold for every N in that range, so most primes in $(N/2, N]$ are m-exceptional whenever

$$m^{1/4} \ll \log N \le \bigl(\varphi(m)/(C\log^2 m)\bigr)^{1/3}. \tag{8}$$

They state this for $N = N(m)$: "In our proof of Theorem 1.1 we actually show that not only is there one exceptional $n > \exp(m^{1/3-\epsilon})$, but that most prime values of n near this bound are exceptions" [PW, p. 2]. The extension to the range (8) is our reading of their proof, checked independently by two internal reviewers against the rendered PDF. Theorem L′ improves the unsimplified bound $\rho_{\mathrm{rep}} \ll L^3\log L\log m/\varphi(m)$ by a factor $(m/\varphi(m))\min(\log m, L/\log m)$, a full $(m/\varphi(m))\log m$ once $L \ge \log^2 m$, which holds near the transition, and extends (8) to $\log N \le \varepsilon(m/\log\log N)^{1/3}$, $\varepsilon \to 0$, with no lower restriction on $\log N$. On the other side, (2) gives "most primes representable" only for $\log N \gg \varphi(m)^{1/2}(\log m)^{3/2}$, and Pomerance and Weingartner wrote that between $\exp(m^{1/3})$ and $\exp(m^{1/2})$ "there is a transition from 'usually false' to 'usually true' " [PW, p. 2]; their Poisson heuristic [PW, p. 3] predicts $m^{1/3+o(1)}$. Theorem U proves the representable half of this prediction at the exact scale $m^{1/3}$, with no logarithmic loss. Pomerance's 2026 talk still lists the location of the transition as open [P].

Elsholtz. The constant $c_{m,3} \asymp m^{-1/3}$ of [E, Rem. 7.3] gives the integer saving $c(\log^2 N/m)^{1/3}$ for $N > N_m$, which exceeds a constant only for $\log N \gg m^{1/2}$. So neither this bound nor (2) reaches the scale $m^{1/3}$, even for integers. Theorem A reaches it for integers (with the loss $(\log m)^{4/3}$ for primes in Corollary D), and Theorem U for primes.

We do not know whether $\rho_{\mathrm{exc}}(m, N)$ is monotone in N. The largest m-exceptional integer, which the heuristic of [PW, p. 3] places near $\exp(m^{1/2})$, is not addressed by any of this.

8.5. **Numerics and a conjecture.** The following is EVIDENCE only. Representability of a prime $p \nmid m$ was decided exactly through [PW, Cor. 2.2, Cor. 2.4] (Type I with $mad \le 3p$, Type II with $mab \le 2p$), validated against an independent brute force for $4 \le m \le 40$, $p < 1500$ (no mismatch), on samples of about 800 primes per window $(N/2, N]$ (`scripts/emn2_scan.c`, `scripts/emn2_half.py`). Let $L_q(m)$ be the value of $\log N$ at which the representable proportion reaches q.

m	60	100	128	200	256	300
$L_{1/2}$	7.64	9.21	9.65	11.21	12.22	13.06
$L_{1/2}/m^{1/3}$	1.95	1.98	1.91	1.92	1.92	1.95
$L_{3/4} - L_{1/4}$	3.73	3.47	4.26	4.83	4.87	5.34

For even $m \in [60, 300]$ one has $L_{1/2}/m^{1/3} = 1.95 \pm 0.05$ with no trend (local log-log slope 0.333 ± 0.02; $L_{1/2}$ is uncertain by ± 0.1–0.2). A factor $(\log m)^{4/3}$ or $(\log m)^{-2/3}$ would change the ratio by 1.56 resp. 0.80 over this range; neither is visible, and within each parity class there is no visible dependence on $\varphi(m)/m$. Odd m sit lower (odd composites 1.77–1.92, odd primes 1.63–1.80) and drift upward (local slopes 0.36 and 0.40); this is a parity effect at these sizes, which the data cannot separate from a slowly varying factor. The window $L_{3/4} - L_{1/4}$, between $0.38L_{1/2}$ and $0.49L_{1/2}$, does not shrink: in A the profile for even m roughly collapses, and

$F(A) = 1 - \exp(-\lambda A^3)$ with $\lambda = \log 2/1.95^3 \approx 0.094$ fits the quartiles, matching the shape $\exp(-(\log p)^3/m)$ of the heuristic of [PW, p. 3]. Solutions are strongly overdispersed (for $m = 200$, $N = 2^{20}$: on average 4.85 Type I/II tuples per prime, yet only 74% of the primes are representable), which reflects the fibre structure of Section 5. The two independent reviews of [MN2] reproduced the half-points with their own decision procedures.

Conjecture C2 (CONJECTURE)**.** *$\rho_{\mathrm{rep}}(m, N) = F(A) + o(1)$ as $m \to \infty$, for a continuous increasing F with $F(0+) = 0$ and $F(\infty) = 1$ (possibly depending on m only through bounded arithmetic data). In particular there is no sharp threshold constant: the transition window has the same order $m^{1/3}$ as its location.*

Theorems U and L′ prove the two tails of C2 up to the gap above. Even the Poisson model with intensity $\lambda L^3/m$ predicts a non-degenerate F; the fibre masses $\mu_c \asymp t^3/m$ of Corollary 5.2 are two-sided only up to constants and the data are overdispersed, so F is not expected to be exactly of Poisson shape.

Assessment 8.11 (why the shapes are what they are)**.** Both our argument and Vaughan's, as reconstructed in [PW, §4], balance a fibre mass μ against a ledger $\mathrm{e}^{O(t\mu)}$, with $t\mu \asymp L = \log N$. With $\mu \asymp t^2/\varphi(m)$, the mass of the forced classes of Vaughan and Pomerance–Weingartner, this gives $\mu \asymp L^{2/3}\varphi(m)^{-1/3}$. With $\mu \asymp t^3/m$ (Corollary 5.2) it gives $\mu \asymp L^{3/4}m^{-1/4} = (L/m^{1/3})^{3/4}$, which is at least one exactly when $L \gg m^{1/3}$. Up to bounded and $\log\log$ factors, the mass t^3/m agrees with the heuristic intensity $(\log^3 p/m)(\log\log p)^{O(1)}$ of [PW, p. 3] and with their rigorous Type II first-moment count. Within the architecture of [TQ], the dependence $m^{-1/4}$ is the mass-driven ceiling: the multiplier range $K = X^{\kappa}$ is already a power of X, and the range $u, v \le x^{1/6}$ is what keeps $z_j^2 < \ell$, which distinctness needs. This mirrors the three-quarter ceiling of [TQ, §10].

9. Under Selberg's eigenvalue conjecture

This section removes the factor $(\log\log N)^{1/3}$ of Section 8.4 *conditionally* on a uniform form of Selberg's eigenvalue conjecture: the transition is then at $\log N \asymp m^{1/3}$ with matching orders on both sides. The source is the internal working document [MN4], which carries the conditional $m = 4$ argument of [HN, Thm. 8.2] over to general m; we state the results and give a proof sketch with exact pointers. [MN4] received one internal hostile review (no fatal or major defect; minor repairs applied), not external refereeing. For a prime $p \nmid m$ let $f_{I,m}(p)$ be the number of Type I solutions of $m/p = 1/x + 1/y + 1/z$, i.e. those with $p \mid x$ and $(p, yz) = 1$, counted with the order of y, z.

Definition 9.1 ((SEL_m))**.** Let $m \ge 4$. *Hypothesis* (SEL_m): for every $d \ge 1$, every squarefree $q \ge 1$ with $(q, 2md) = 1$ and every even Dirichlet character χ modulo q, the Laplacian $-\Delta$ on $L^2(\Gamma_0(mdq^2)\backslash\mathbb{H}, \chi)$ has no eigenvalue in $(0, 1/4)$.

Each character space embeds in $L^2(\Gamma_1(mdq^2)\backslash\mathbb{H})$, so (SEL_m) for all m follows from Selberg's conjecture $\lambda_1 \ge 1/4$ for every $\Gamma_1(M)$. For $m = 4$ it is the hypothesis (SEL) of [HN, Hyp. 1.1] (levels $4dq^2$, q odd squarefree, $(q, d) = 1$, even characters modulo $2q$, equivalently modulo q). It is used only for levels mdq^2 with $m \le (\log N)^5$.

Theorem 9.2 (CONDITIONAL on (SEL_m); proved in [MN4] relative to the inputs of [HN, Thm. 8.2] and [MN3])**.** *There is an absolute N_0 such that for $N \ge N_0$,*

$L = \log N$ and $4 \le m \le L^5$,

$$\sum_{N/2<p\le N} f_{I,m}(p) \ll \frac{N(L^2 + L\log^2 m)}{m} + \frac{N\, m^{-0.35}}{L}.$$

Theorem 9.3 (Theorem L″)**.** *For all $m \ge 4$ and $N \ge 16$, with $L = \log N$ and $A = L/m^{1/3}$,*

$$\rho_{\mathrm{rep}}(m,N) \ll \frac{L^3 + L^2\log^2 m}{m} + m^{-0.35} = A^3 + A^2 m^{-1/3}\log^2 m + m^{-0.35}.$$

This is CONDITIONAL *on (SEL$_m$) when $m \le L^5$, and* PROVED *(unconditionally, from Theorem L′ and Lemma 8.9) when $m > L^5$.*

Corollary 9.4 (sharp order of the transition)**.** *For every $\varepsilon \in (0,1)$ there are $c_\varepsilon, A_\varepsilon > 0$ and m_ε such that for all $m \ge m_\varepsilon$ and $N \ge 16$:*

(a) (CONDITIONAL *on (SEL$_m$)*) *if $\log N \le c_\varepsilon m^{1/3}$ then $\rho_{\mathrm{rep}}(m,N) \le \varepsilon$;*

(b) (PROVED*, Theorem U; ineffective*) *if $\log N \ge A_\varepsilon m^{1/3}$ then $\rho_{\mathrm{exc}}(m,N) \le \varepsilon$.*

So under (SEL$_m$) the density transition for m/p lies at $\log N \asymp m^{1/3}$, with matching orders on both sides.

Proof. (a) By Theorem 9.3 with $A \le c_\varepsilon$, $\rho_{\mathrm{rep}} \le C(c_\varepsilon^3 + c_\varepsilon^2 m^{-1/3}\log^2 m + m^{-0.35})$, which is $\le \varepsilon$ for $c_\varepsilon = (\varepsilon/3C)^{1/3}$ and $m \ge m_\varepsilon$. (b) is Theorem U, valid for all $m \ge 4$. □

The restriction $m \ge m_\varepsilon$ only makes $m^{-0.35}$ and $m^{-1/3}\log^2 m$ small; for bounded m the transition concerns bounded N. Theorem 9.3 gives the profile bound $\rho_{\mathrm{rep}} \ll A^3$ for $m^{-0.11} \ll A \ll 1$, the shape of the small-A end of the fit of Section 8.5 (which is EVIDENCE only). Nothing here bears on Conjecture C2: Corollary 9.4 fixes the order of the transition, not the existence of a limit profile $F(A)$.

Proof of Theorem 9.3, given Theorem 9.2. This is [MN4, Thm. 5.1]. (i) If $\log m > L/10$, Lemma 8.9 gives $\rho_{\mathrm{rep}} \le \mathrm{e}^{CL/\log L}/m$. For $L \ge L_0 := \mathrm{e}^{20C}$ we have $CL/\log L \le 10C\log m/\log L \le 0.65\log m$, so $\rho_{\mathrm{rep}} \le m^{-0.35}$. For $L < L_0$ either $\rho_{\mathrm{rep}} = 0$ or $m \le 3\mathrm{e}^L < 3\mathrm{e}^{L_0}$, and the claim is trivial. (ii) If $\log m \le L/10$ and $m > L^5$, then $L \le m^{1/5}$, so $L^3\log L/m \le m^{-2/5}\log m$ and $L^2\log^2 m \log L/m \le m^{-3/5}\log^3 m$, both $\ll m^{-0.35}$; Theorem L′ (whose hypothesis $L \le m^{1/2}$ holds) gives $\rho_{\mathrm{rep}} \ll m^{-0.35}$, unconditionally. (iii) If $m \le L^5$, then $\log m \le L/10$ for $N \ge N_0$. The primes with a Type II solution number $\ll (N/L)(L^3 + m^{0.02})/m$ (Proposition 8.5), and those with a Type I solution are at most $\sum_{N/2<p\le N} f_{I,m}(p)$, bounded by Theorem 9.2. Divide by $\pi^*(N) \gg N/L$ and use $m^{0.02}/m \le m^{-0.35}$. For $N < N_0$ the integer $m \le L^5$ is bounded and the claim is trivial. □

Sketch of proof of Theorem 9.2. The full argument is [MN4, §§1–4], which runs the proof of [HN, Thm. 8.2] (the case $m = 4$ over $p \le N$) with 4 replaced by m. We list the steps and what changes.

Why $m \le L^5$ costs nothing. For $m > L^5$, Theorem L′ already gives $\rho_{\mathrm{rep}} \ll m^{-0.35}$ (part (ii) above; [MN4, Lemma 0.1]). For $m \le L^5$ every factor $m^{O(1)}$ in a *remainder* is $L^{O(1)}$, and the proof of [HN, Thm. 8.2] already absorbs $L^{O(1)}$ losses in remainders: the layers with saving parameter $k \le C_1 \log L$ go to Brun–Titchmarsh.

Reduction ([MN4, Lemma 1.1], PROVED). The proof of [ET, Prop. 2.2] with 4 replaced by m gives $f_{I,m}(n) \le 2\sum_{c\ge 1} w_{c,m}(n)$, where $w_{c,m}(n)$ counts (a,d,f) with

$f \mid ma^2d+1$, $n = macd - f$, $0 < f \le 2n$. Since $ef = ma^2d+1$, the variables e, f, n are coprime to m and n is coprime to d; so no prime dividing md can be sieved.

Heegner forms ([MN4, §2.1], PROVED). For fixed $t = md$ the forms $[f, 2ta, te]$ with $ef - ta^2 = 1$ are exactly the positive definite forms $[U, B, V]$ of discriminant $-4t$ with $2t \mid B$, $t \mid V$. This set is $\Gamma^0(t)$-stable, with no parity condition, and its roots satisfy $\cosh \operatorname{dist} \ge 3/2$, uniformly in m, by [MN4, Lemma 2.2_m] (cf. [HN, Lemma 3.2]). The level $4dq^2$ of [HN] becomes $M = mdq^2$, and the group $\Gamma(2q)$ becomes $\Gamma(q)$.

Spectral side ([MN4, §2.2], CONDITIONAL on (SEL_m)). The box-variance bound [HN, Prop. 5.1] holds verbatim for the groups $\Gamma_0(mdq^2)$ with even nebentypus modulo q (the infinity cusp has width 1; elliptic elements, which occur for small md, are harmless because [HN] uses orbifold weights). This is the only place where (SEL_m) is used. It gives the per-d count [MN4, Thm. 6.2_m], the analogue of [HN, Thm. 6.2], with local densities computed in [MN4, Lemma 6.1_m]. The per-a Poisson–Weil count [HN, Prop. 7.1] carries over with modulus ma^2 [MN4, §2.6]. Its saving and that of the per-d count again degenerate linearly at $d = a$. The value of m enters the remainders only polynomially, as $(md/A)^{1/2}$ and $m^{1/2}A/D$.

Main terms ([MN4, §3], PROVED). Since the sieve cannot use the primes $\ell \mid m$, every sieve product loses $m/\varphi(m)$. This is paid for by the coprimality gain $\varphi(m)/m$ of the masses: the divisors e, f of ma^2d+1 are coprime to m (Lemma 8.6, i.e. [MN3, Prop. 2.3]). A Brun–Titchmarsh sum over moduli mx alone has no such gain, so each one is arranged to take its gain from a variable coprime to m. One weighted divisor-sum estimate, [MN4, §3.2(c)], holds only for $D, F \ge L^{100}$ and is used only there; the range $D \le L^{100}$ is handled separately by aggregated Brun–Titchmarsh. The large-c part, [MN4, §3.4], contributes the term $L \log^2 m$ and the term $m^{-0.35}$.

Assembly ([MN4, Thm. 4.1]). Steps (1)–(5) (as in [TTL]) of the proof of [HN, Thm. 8.2], plus $D \le L^{100}$, hold with every main term multiplied by $1/m$ and with no factor $m/\varphi(m)$. The degenerating savings are summed by [HN, Lemma 8.1]. □

Remark 9.5 (what is unconditional for general m)**.** The unconditional parts of [MN4] are the reduction, the separation and invariance of the level-md Heegner forms, the local densities, the class counts and the per-a count, and the Brun–Titchmarsh masses with the exact gain $\varphi(m)/m$ [MN4, §6]. Without (SEL_m) the lower side is still Theorem L′, and the transition is known only within $c(m/\log m)^{1/3} \le \log N \le A\, m^{1/3}$. With the single-level Kim–Sarnak bound the argument leaves a strip of positive width next to $d = a$, as in [HN, §10], independently of m. For $m = 4$, [HN, Thm. 1] (= [HN, Thm. 9.9(i)]) removes this obstruction *unconditionally*, by averaging the exceptional spectrum over the levels $4dq^2$, $d \asymp D$, with a level-averaged large sieve of Deshouillers and Iwaniec in Drappeau's nebentypus form [HN, §9]. This gives $\sum_{p \le N} f_I(p) = o(N \log^2 N \log\log N)$ without a rate; the bound $\ll N \log^2 N$ [HN, Thm. 9.9(ii)] is conditional on a quantified form (EFF) of the ε-constants in those estimates. Neither [MN4] nor [HN] treats the levels mdq^2 for general m by this averaging, so whether the unconditional result transfers, uniformly for $m \le L^5$, is *open* (Problem 2). Even an m-uniform transfer of [HN, Thm. 1] would only improve Theorem L′ by an unquantified factor $o(1)$, not give the exact scale. An m-uniform transfer of the (EFF) case would give Corollary 9.4(a) conditionally on (EFF) instead of (SEL_m).

10. The lower end of the window range

Proposition E. *Let $m \geq 4$. Suppose that for some $x_0, C, c > 0$ and some function $H_0(x) \geq 1$ with $CH_0(x)\mathrm{e}^{-c(\log x)^{3/4}} < 1$ for all real $x \geq x_0$ one has*

$$E_m((x, x + H_0(x)]) \leq CH_0(x)\mathrm{e}^{-c(\log x)^{3/4}} \qquad \text{for all real } x \geq x_0.$$

Then there is no m-exceptional $n > x_0$.

Proof. Each such window contains fewer than one, hence no, exceptional integer. Every integer $n > x_0$ lies in one of them: take $x = n - 1$ if $n - 1 \geq x_0$, and $x = x_0$ otherwise, using $H_0 \geq 1$. □

Thus the target shape $E_m((x, x + H]) \ll H\mathrm{e}^{-c(\log x)^{3/4}}$ cannot be proved for any $H(x)$ with $CH(x) < \mathrm{e}^{c(\log x)^{3/4}}$, for instance for polylogarithmic H, without proving that all large n are m-representable. More generally, a bound $CH\mathrm{e}^{-S}$ with $S > \log(CH)$ on a window of length H forces it to be free of exceptions. On the scale $S = c(\log H)^\theta$, any $\theta > 1$, or $\theta = 1$ with $c > 1$, is the full conjecture on long windows, while $\theta \leq 1$ with small c is not.

Define the *window exponent* $\theta_{\mathrm{win}}(m)$ as the supremum of the θ for which $E_m(I) \ll_m |I| \exp\{-c(\log |I|)^\theta\}$ holds for some $c > 0$, uniformly over all intervals I with $|I| \geq 2$.

Proposition 10.1. (a) $\theta_{\mathrm{win}}(m) \geq 3/4$ *for every* $m \geq 4$. (b) *If* $\theta_{\mathrm{win}}(m) > 3/4$, *then* $E_m(N) \ll N \exp\{-c(\log N)^\theta\}$ *for some* $\theta > 3/4$.

Proof. (a) is Theorem A; (b) is the case $I = (0, N]$. □

So the window problem is at least as hard as the problem over initial segments. The remaining statements of this section are heuristic.

Assessment 10.2 (three regimes)**.** For windows $(x, x + H]$ and fixed m: if $H \geq x^\theta$, Theorem A gives saving $\geq c_\theta(\log x)^{3/4}$; if $\mathrm{e}^{c(\log x)^{3/4}} \lesssim H \leq x^{o(1)}$, it gives only $(\log H)^{3/4} = o((\log x)^{3/4})$, and a saving of order $(\log x)^{3/4}$ would need either $\theta_{\mathrm{win}} > 3/4$ or input that depends on the position of the window; if $CH < \mathrm{e}^{c(\log x)^{3/4}}$, such a saving is equivalent to having no exception in the window (Proposition E).

Assessment 10.3 (shift-uniform methods)**.** Lemma 7.1 is shift-uniform: it bounds the maximum over z of the number of integers in $(z, z + H]$ that avoid every class of the family. Any argument that uses only "exceptional n avoid the classes of a fixed family $\mathfrak{G}$" and the window length is bounded below by this maximum $M_{\mathfrak{G}}(H)$. For such methods, the short-window bound at length H is controlled by the same quantity as the bound over $(0, H]$; it is not equal to it, since $(0, H]$ is one shift among many. Whether $(\log H)^{3/4}$ is optimal for such methods is open. We know of no Jacobsthal- or Maier-type construction of windows of length H containing $\geq H\mathrm{e}^{-C(\log H)^{3/4}}$ avoiders. Concerning position: the family used at length H has $X = \exp\{\alpha(m \log H)^{1/4}\}$ and period $\mathcal{M} = \mathrm{e}^{O(X)}$. If $\mathcal{M} + H \leq x$, which holds for $H \leq \exp\{c(\log\log x)^4/m\}$ with a suitable c, every translate class of the period occurs inside $[x, 2x]$. So for that family the worst window lies in $[x, 2x]$, and the position cannot help a method that uses only this family and the window length. To go beyond $(\log H)^{3/4}$ in $(x, x + H]$ with $H = x^{o(1)}$, a method would have to use something else: rounding effects of the counts of classes of modulus $q \leq H$ at the specific position, correlations between their intersections, or classes of modulus $> H$,

each of which contains at most one point of the window. Nothing here excludes such input.

11. Literature

The following was checked in October 2026 by web searches and by reading the papers where accessible. It is an assessment, not an exhaustive survey.

- Elsholtz and Tao [ET, §1] cite Vaughan's bound $N\exp\{-c(\log N)^{2/3}\}$ over $n < N$ ("for instance"), and refer to Li Delang [Li] and Yang [Y], among others, for weaker results. Yang's abstract concerns the initial-segment count; we did not read Li's or Yang's papers in full. There is no short-interval or progression statement in [ET].
- Pomerance and Weingartner [PW] describe Vaughan's 1970 bound as the state of the art for $4/n$ [PW, p. 2], and prove (2) for general m, again over $n \le N$. The abstract of the cited version (v2) says: "A result of Vaughan is that for each m, most n's have m/n representable; we make the dependence on m in this result explicit." (The first arXiv version said instead that they "generalize a result of Vaughan".) So, according to [PW], Vaughan's result already covers each fixed m. We could not read Vaughan's paper and rely on this secondary statement for the dependence on m. Elsholtz [E, Rem. 7.3] gives the admissible constant $c_{m,3} = 3e^{-2/3}(8m)^{-1/3} - \varepsilon$ in Vaughan's bound for $N > N_m$ (with $c_{4,3} = 0.5645$); [PW, Thm. 1.3] is the first bound *uniform* in m that we can cite (O101 audit).
- Elsholtz [E, p. 3210] proves $E_{m,k}(N) \ll N\exp\{-c_{m,k}(\log N)^{1-1/(2^{k-1}-1)}\}$ for $k \ge 3$ unit fractions, improving Viola [Vi] (exponent $1 - 1/(k-1)$) and Shen [Sh] (exponent $1 - 1/k$). For $k = 3$ none of these exceeds Vaughan's $2/3$. Sander [Sa] gives, for a single Type II subfamily, a lower bound $\gg x/(\log x)^{3/2}$ for the primes in a progression on which that subfamily fails. This is the only progression-restricted sieve result for $4/n$ that we found, and it is not an exceptional-set bound (O101 audit; [Vi, Sh, Sa] only via secondary sources).
- Searches for short-interval, almost-all-short-interval and progression versions of exceptional-set bounds for $4/n$ or m/n, for "Sierpiński $5/n$ exceptional set" and for "$(\log N)^{3/4}$" returned nothing relevant in October 2026; these searches missed the preprint [KRK], which a later (November 2026) audit located and which has the exponent $3/4$ over initial segments (see below). Verification papers (to 10^{17} and 10^{18}) concern computation, not density.
- Bibliographic status: [ET], [PW] (arXiv v2) and [S] were consulted in full text. The entries [Li] and [Y] were checked only against the bibliography of [ET]; [V] was not accessible to us; [MV], [D] and [BHP] are standard references that we did not re-read for this note.

We found no published short-interval or progression version. The exponent $3/4$ over initial segments was claimed earlier by Kandukoori, Ramesh and Kumar [KRK], for $m = 4$ (their Theorem 1) and for each fixed $m \ge 4$ with constants depending on m (their Theorem 2); their preprint is unrefereed and their analytic proof was not re-audited here, and the internal argument [TQ] is likewise not externally refereed. As far as we can tell, the short-interval and progression forms of Theorem A, its uniform dependence on m, Corollary D and Theorems U and L′ are new relative to [PW], [E, Rem. 7.3] and the stated theorems of [KRK]; the location of the transition is posed

as open in [PW, p. 2] and [P]. The method is that of [TQ] with 4 replaced by m. As explained in the introduction, a specialist would expect the short-interval transfer, and presumably Vaughan's argument already gives $E_4(I) \ll |I| \exp\{-c(\log|I|)^{2/3}\}$ for intervals. An earlier internal document observed the shift-uniformity of the 3/4 majorant for large primes [W]; Lemma 7.3 extends this to all integers.

12. Open problems

Problem 1 (large smooth moduli)**.** Remove the factor q_1 in Theorem B, at least for $(b, q_1) = 1$. For such b, conditioning on $n \equiv b\ (q_1)$ forces $p \nmid c$ for every $p \mid q_1$, which only makes the bad event $\{Z(c) > 1/4\}$ less likely. Fixing the coordinates $n \bmod \ell$ for $\ell \mid q_1$ changes the fibre product by a factor $\le \exp\{\sum_{\ell \mid q_1,\, \ell > X^{1/2}} 2\ell^{-2/3}\} \le \exp\{2\omega(q_1)X^{-1/3}\}$, using $f_c(\ell) \le z_j^2 \le \ell^{1/3}$; this is $1 + o(1)$ when $\omega(q_1) = o(X^{1/3})$, but not for every q_1. What is missing is a version of Lemma 6.2 conditional on $n \equiv b\ (q_1)$. This is probably routine but has not been attempted. A related question is whether every prime modulus q is covered for general m (Remark 7.5).

Problem 2 (the transition gap)**.** Close the gap $(\log\log N)^{1/3}$ between Theorems L′ and U unconditionally. One loss remains, on the lower side: (i) a Type I prime count without the Brun–Titchmarsh $\log L$, uniformly in m; this is the open $\log\log N$ of [ET, Thm. 1.1] already for $m = 4$. It is removed conditionally on Selberg's eigenvalue conjecture, uniformly for $m \le L^5$ (Theorem 9.2, Corollary 9.4); for $m = 4$ it is reduced unconditionally to $o(\log\log N)$ without a rate [HN, Thm. 1]. Does the level-averaged argument of [HN, §9] carry over to the levels mdq^2, uniformly for $m \le L^5$ (Remark 9.5)? With a rate, i.e. with an m-uniform form of [HN, Thm. 9.9(ii)], this would make Corollary 9.4(a) conditional only on (EFF). (The factor $m/\varphi(m)$ of [MN2] is removed by Lemma 8.6.) Also: (ii) an effective, quantitative form of Theorem U in the range $1 \ll A \ll (\log m)^{4/3}$ not covered by Corollary D. Is $\rho_{\mathrm{exc}}(m, N)$ monotone in N? For $m \ge \varepsilon(\log N)^3$ Theorem A is trivial, and when $(\log N)^3 \log\log N/m \to 0$ most primes are exceptional (Theorem L′), so $E_m(N) \gg N/\log N$ there and no saving beyond $\log\log N + O(1)$ is possible. What is the true size of $E_m(N)/N$ in between?

Problem 3 (the profile)**.** Prove or disprove Conjecture C2: is $\rho_{\mathrm{rep}}(m, N) = F(A) + o(1)$ with a non-degenerate F, so that there is no sharp threshold constant? Explain the parity effect of Section 8.5.

Problem 4 (position-dependent input)**.** Determine $\theta_{\mathrm{win}}(m)$. Equivalently for the methods of Assessment 10.3: are there windows of length H that contain $\ge H\mathrm{e}^{-C(\log H)^{3/4}}$ integers avoiding the forced classes? If not, what position-dependent input could improve on $(\log H)^{3/4}$ in $(x, x + H]$ for $H = x^{o(1)}$?

Problem 5 (the exponent)**.** Improve the exponent 3/4 itself, for any m. The ceiling discussion of [TQ, §10] and Assessment 8.11 indicate that this needs a new source of class mass, not a better transfer.

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